METRIC

MIL-S-62546A(AT)
11 August 1989
SUPERSEDING
(see 6.4)

MILITARY SPECIFICATION

SENSOR, FIRE, OPTICAL

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers two types of optical fire sensor assemblies (OFSAs), components of the Halon 1301 automatic fire extinguishing systems used in military vehicles. The OFSA detects and signals the presence of a hydrocarbon fire requiring a system response (see 6.1).

1.2 Classification. OFSAs shall be of the following types, as specified (see 6.2 and 6.5):

Type I	- Discriminatory
Type II	- Non-discriminatory

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-B-18/9	- Battery, Dry, BA-30.
MIL-F-3747	- Flashlight, Plastic Case, Tubular (Regular Explosion Proof, Explosion Proof Heat Resistant, Traffic Directing, and Inspection Light).
MIL-F-13927	- Fungus Resistance Tests: Automotive Components
MIL-P-40061	- Panel Marker, Aerial Liaison, VS-17/GVX
MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire Resistant Syntnetic Hydrocarbon Base.

STANDARDS

MILITARY

MIL-STD-130	- Identification Markings of US Military Property
MIL-STD-171	- Finishing of Metal and Wood Surfaces.
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurements of
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-1275	- Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.

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MS3114	- Connectors, Receptacle, Electric, Solder Type, Rear Mounting Jam Nut, Bayonet Coupling.
MS3121	- Connectors, Plug, Electric, Crimp Type, Cable Connecting, Bayonet Coupling
MS3449	- Connectors, Receptacle, Electrical, Series 2, Single Hole Mount, Bayonet Coupling, Solder Pin Contact, Class H.
MS3474	- Connectors, Receptacle, Electrical, Series 2, Crimp Type, Narrow Flange Mount, Bayonet Coupling, Classes A, L, S and W.
MS51073	- Light Dome, 24 Volt DC.
MS53023	- Headlight and Service, Double Filament, 24 Volt, Waterproof.
MS53024	- Headlight and Infrared, Double Filament, 24 Volt, Waterproof.

HANDBOOKS

MILITARY

MIL-HDBK-217	- Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ARMY

12370458	- Optical Fire Sensor Assembly.
12370459	- Module, Standard Control Electronic.
12343265	- Connector, Electrical Receptacle

PUBLICATIONS

TEST PROCEDURES

Nuclear Survivability Criteria	- Nuclear Survivability Criteria for the Automatic Fire Extinguishing System (AFES) (U). Document is classified confidential.

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(Copies of drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity [see 6.6]).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. OFSAs furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) at the time set for opening of bids (see 4.4 and 6.3).

3.2 Materials. Materials shall be as specified herein and as specified in referenced specifications, standards and drawings. Materials shall be free of defects which adversely affect performance or serviceability of the OFSAs. Materials shall be suitable for use over an ambient temperature range from minus (-) 51 to 125 degrees Celsius (°C) [-60 to 257 degrees Fahrenheit (°F)]. There shall be no changes in physical or dimensional characteristics that could result in the operation of the OFSA falling outside the limits specified herein (see 4.7.1).

3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which could encourage galvanic action, shall be avoided. Separation of dissimilar metals shall be accomplished by providing approved protective finishes, coatings or insulation between the mating materials and shall be in accordance with provisions of MIL-STD-171 which address reduction of corrosion at intermetallic contact points. Corrosion-resistant materials shall be used whenever possible (see 4.7.1).

3.2.2 Encapsulation. Encapsulation, if used, shall be of silicone rubber compounds to permit removal for repair (see 4.7.1).

3.2.3 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.5.5).

3.3 Design and construction. The OFSA shall be designed to function in a military vehicle automatic fire extinguishing system as specified in MIL-S-62544. The OFSA shall optically detect and signal the presence of a hydrocarbon fire. The OFSA shall provide signals in response to the built-in-test-equipment (BITE), indicating its functional condition. The OFSA shall operate in a temperature range of -51 to 125°C (-60 to 257°F) (see 4.7.1).

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3.3.1 Electrical interface. Electrical interface shall be provided through the use of a connector conforming to MS3474, W12-8PN, MS3449, H12-8PN; MS3114, W12-8PN; MS3121, W12-8PN; or 12343265-1 Pin assignments shall be as specified on Drawing 12370458 and table I (see 4.7.1, 4.7.2 and 4.7.3).

3.3.2 Configuration The OFSA shall conform to the physical and dimensional requirements of Drawing 12370458 (see 4.7.1 and 4.7.2)

3.3.3 Vehicle armor type coding. The OFSA shall be capable of operating in either a steel or aluminum armored vehicle. The vehicle wiring harness will provide coded information as to which type of armor by way of two connector pins at each OFSA. The two designated pins (see table I) will be connected within the harness for aluminum vehicles and will not be used for steel vehicles This coded information may be used by the OFSA manufacturer to modify sensor logic for steel or aluminum applications. If the OFSA logic is independent of armor type, the two code pins shall not be used (see 4.7.3)

3.3.4 BITE. The OFSA shall conduct a total electronic check (see 3.4.1.3) when it receives a BITE activation signal on pin E (see table I). Each OFSA shall generate optical stimuli causing the OFSA to generate both a small fire and a large fire signal. The stimuli source may be external to the OFSA protective window(s) A check for dirty sensor windows shall be included (see 4.7.3).

3.3.5 Reliability. The OFSA's reliability shall be determined on the basis of mean time between failure (MTBF) as calculated in accordance with MIL-HDBK-217 The OFSA shall have not less than 100,000 operating hours between failures A failure shall be defined as any OFSA malfunction which prevents the component from responding as specified (see 4.7.4).

3.3.6 Voltage characteristics The OFSA shall meet the requirements of MIL-STD-1275 without loss of effectiveness and without causing damage or malfunction to interfacing components. The OFSA shall not false alarm before, during, or after testing (see 4.7.3).

3.3.7 Qualified products Any component of the OFSA, when listed on a QPL, shall be furnished from a source listed or approved for listing on the QPL in accordance with the applicable specification or drawing at the time set for opening of bids Documentation of the item acquisition from a qualified source shall be made available to the Government for review The documentation shall include the QPL date, identification of the supplier, purchase order and quantity (see 4.1.3 and 4.7.1).

3.4 Performance. The OFSA shall be capable of detecting hydrocarbon mist or surface fires, differentiating between large and small fires and providing an appropriate output response. Response to a large or small fire shall be an output signal A at connector pin C and/or D in accordance with table I Specific response requirements are given in subsequent paragraphs. Shown for reference in figure 1 is a characteristic optical emission spectrum

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TABLE I. OFSA connector J3. 1/

PIN	FUNCTION	In/Out	Signal (See table I)	Duration
A	POWER.	In	16 to 30 V dc in accordance with MIL-STD-1275	
B	RETURN	In		
C	LARGE FIRE SIGNAL. Produced when (1) Sensing a fire which has reached the sensitivity threshold (defined in 3.4.1.2.2), or (2) Following receipt of BITE signal (Pin E).	Out	Signal A 2/	(1) As long as fire exceeds threshold, or (2) For less than 5 ms and not more than 4.0s following receipt of BITE signal (Pin E).
D	SMALL FIRE SIGNAL. Produced when (1) Sensing a fire which has reached the sensitivity threshold (defined in 3.4.1.2.1), or (2) Following receipt of BITE signal (Pin E).	Out	Signal A	(1) As long as fire exceeds threshold, or (2) For less than 5 ms and not more than 4.0s following receipt of BITE signal (Pin E).
E	BITE SIGNAL. A signal received at this pin shall initiate a test of OFSA functions (as described in 3.4.1.3) and shall result in output signals from Pins C and D.	In	Signal A	5 ms to 4.0 s
F and G	CODE PINS. These pins are connected and together within the wiring harness for vehicles with aluminum armor. With steel armor, the pins are not used (see 3.3.3).			
H	UNUSED			

1/ OFSA connector (see 3.3.1).

2/ Signal A: 10 to 30 V dc, source output impedance less than or equal to (≤) 2000 ohms, load output impedance ≤ 20 000 ohms; Quiescent conditions: 1 V dc maximum, input impedance (source and load) greater than or equal to (≤) 20 000 ohms.

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for an open diesel fuel (DF-2 per VV-F-800) fire. Figure 2 is a characteristic optical emission spectrum for an open hydraulic fluid (MIL-H-46170) fire. These spectra should be considered typical of optical emissions for which the OFSA shall respond (see 4.7.5).

3.4.1 Response characteristics. The OFSA shall respond to radiation stimuli as indicated in the following paragraphs (see 4.7.5.1).

3.4.1.1 Field of view (FOV). The FOV of the OFSA shall be, as a minimum, a 90 degree (°) solid cone. Anywhere within the FOV the OFSA shall meet the response characteristic requirements specified herein (see 4.7.5.1.1).

3.4.1.2 Thresholds. The OFSA shall respond to two discrete energy threshold levels: A lower energy threshold, small fire; and a higher energy threshold, large fire. The response shall be in the form of signal outputs on the appropriate connector pins in accordance with table I (see 4.7.5.1.2).

3.4.1.2.1 Small fire. The OFSA shall generate output signal "A" (see table I) on pin D in not more than 5 seconds (s) when exposed to a pan-fire, 130 millimeters (mm) [5 inches (in)] in diameter and 76 mm (3 in) deep containing 840 cubic centimeters (cm^3) [51 cubic inches (in^3)] of diesel fuel, DF-2 per VV-F-800) at a distance of 1200 mm (48 in). No large fire signal shall be generated on pin C (see 4.7.5.1.2.1 and figure 3).

3.4.1.2.2 Large fire. The OFSA shall generate output signal "A" (see table I) on pins C and D in not more than 5 s to the hydrocarbon pan-fire described in 3.4.1.2.1, except the distance shall be 380 mm (15 in) from the centerline of the pan to the OFSA (see 4.7.5.1.2.2).

3.4.1.3 BITE. Upon receipt of BITE signal "A" (see table I) on pin E the OFSA shall generate optical stimuli to perform a complete functional self test (see 3.3.4). If the OFSA is functioning properly and if the optical window(s) does not require cleaning, it shall cause simultaneous "A" signals on output pins C and D within 5 s. If the OFSA is not functioning properly or requires window cleaning, it shall not generate output signals on those pins (see 4.7.5.1.3).

3.4.1.4 Sensitivity. The OFSA shall respond as required in 3.4.1.1 through 3.4.1.3 (see 4.7.5.1.4):

a. When the OFSA optical lens or window(s) is facing directly into the sun (see 4.7.5.1.4.1).
b. After diesel fuel immersion (see 4.7.5.1.4.2).

3.4.1.5 Response time. At all temperatures between -17 and 71°C (1 and 160°F), the OFSA shall respond to the radiation from a 250-mm (10 in) diameter pan-fire, 76 mm (3 in) deep containing 3100 cm^3 (188 in^3) of diesel fuel (DF-2 per VV-F-800) at a distance of 380 mm (15 in) from the OFSA and shall provide the appropriate signal(s) within the following times (see 4.7.5.1.5).

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	Time (ms)
a. Type I OFSA	4
b. Type II OFSA	3

3.4.1.6 Penetration response time. The OFSA shall respond within the time specified in 3.4.1.5, to a hydrocarbon fire ignited subsequent to and resulting from penetration of steel and aluminum armor by threat munitions. Conditions shall be in accordance with 4.7.5.2.2 with the OFSA located not less than 380 mm (15 in) from the fire (see 4.7.5.2.2.4).

3.4.2 False alarm susceptibility. The OFSA shall respond to radiation stimuli as indicated in the following paragraphs (see 4.7.5.2).

3.4.2.1 Radiation stimuli. The OFSA shall not respond at distances equal to or greater than the immunity distance when exposed to the radiation sources of table II throughout the FOV of the OFSA (except for chopped light requirements). This requirement shall be met with the system on, source turned on and off; source on, system turned on and off; (see 4.7.5.2.1.15) and chopped light, as specified herein (see 4.7.5.2.1).

TABLE II. False alarm susceptibility

Requirement 3.4.2.1 step	Test 4.7.5.2.1	Radiation source description	Immunity distance [mm(in)]	
			Small fire	Large fire
1	4.7.5.2.1.1	Vehicle headlight (low beam) conforming to MS53023-1	300 (12)	100 (4)
2	4.7.5.2.1.2 4.7.5.2.1.3 4.7.5.2.1.5	Sunlight	IAD 1/	IAD
3	Step 1, table VII	Incandescent frosted glass light, 100 W	150 (6)	25 (1)
4	Step 2, table VII	Incandescent clear glass light, rough service, 100 W	225 (9)	50 (2)
5	Step 3, table VII	Fluorescent light with white enamel reflector, standard office or shop, 40 W (or two 20 W)	150 (6)	IAD
6	Step 7, table VII	Electric arc [12 mm (15/32 in) gap at 4000 V alternating current, 60 Hz]	25 (1)	25 (1)

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TABLE II. False alarm susceptibility - Continued.

Requirement 3.4.2.1 step	Test 4.7.5.2.1	Radiation source description	Immunity distance [mm(in)] Small fire	Large fire
7	Step 4, table VII	Vehicle IP light conforming (low beam) to MS53024-1	600 (24)	300 (12)
8	4.7.5.2.1.3	Ambient light extremes (vehicle darkness to bright light with snow, water, rain, desert glare and fog)	IAD	IAD
-	4.7.5.2.1.4 4.7.5.2.1.5	Bright colored clothing, including red and safety orange at 1500 mm (60 in) and to near zero	IAD	IAD
10	Step 8, table VII	Electronic flash (180 watt-seconds minimum output)	450 (18)	225 (9)
11	Step 9, table VII	Movie light, 625 W quartz DWY lamp (Sylvania S.G.-55 or equivalent)	1200 (48)	600 (24)
12	Step 5, table VII	Red dome light conforming to MS51073-1	IAD	IAD
13	Step 6, table VII	Blue-green dome light conforming to MS51073-1	IAD	IAD
14	4.7.5.2.1.6	Flashlight (MX 991/U)	IAD	IAD
15	4.7.5.2.1.7	Indirect or reflected sunlight	IAD	IAD
16	4.7.5.2.1.1 4.7.5.2.1.2 4.7.5.2.1.5 4.7.5.2.1.6 4.7.5.2.1.8	Chopped light - individual sources	2/	2/
17	4.7.5.2.1.3 4.7.5.2.1.9	Chopped light - combination of sources	3/	3/
18	4.7.5.2.1.10	Radiation heater, 1500 W	900 (36)	450 (18)

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TABLE II. False alarm susceptibility - Continued.

Requirement 3.4.2.1 step	Test 4.7.5.2.1	Radiation source description	Immunity distance [mm (in)]	
			Small fire	Large fire
19	4.7.5.2.1.10	Radiation heater, 1000 W with fan	600 (24)	300 (12)
20	4.7.5.2.1.11	Arc welding [4 mm (5/32 in) rod; 300 A]	1500 (60)	300 (12)
21	4.7.5.2.1.11	Acetylene welding [00 tip, 16 x 150 mm (5/8 by 6 in) flame]	1500 (60)	300 (12)
22	4.7.5.2.1.12a	Rifle flash from M16 rifle	250 (10)	50 (2)
23	4.7.5.2.1.12b	Muzzle flash from 105-mm (4-in) gun	4/	4/
24	4.7.5.2.1.12c	Muzzle flash from 50-caliber machine gun	4/	4/
25	4.7.5.2.1.13	Lit cigar	100 (4)	25 (1)
26	4.7.5.2.1.13	Lit cigarette	100 (4)	25 (1)
27	4.7.5.2.1.14	Match, wood, stick including flare up	300 (12)	100 (4)
28	4.7.5.2.1.14	Match, paper, including flare up	200 (8)	100 (4)

1/ Immune at any distance (IAD). Includes the case when stimuli is physically as near the sensor as possible
2/ To be the same immunity distance as when not chopped.
3/ The immunity distance of the combination indicated when not chopped, for instance, 300 and 50 mm (12 and 2 in), the immunity distance is 300 mm (12 in) for combination.
4/ Distances to be measured and recorded

3.4.2.2 Penetrating munitions. Type I OFSAs shall not respond to the high intensity, short duration emissions (in relation to hydrocarbon fires) caused by munition penetration of aluminum and steel armors. The OFSA shall demonstrate a minimum discrimination level (not generate large fire signals on pin C) as shown in table III (see 4.7.5.2.2)

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TABLE III. Discrimination performance.

Armor	Munition	Discrimination Requirement
32 mm (1.25 in) aluminum	25 mm (1 in) kinetic energy	90%
51 mm (2 in) steel	38 mm (1.5 in) chemical energy	90%
32 mm (1.25 in) steel applique over 25 mm (1 in) aluminum with spall liners	89 mm (3.5 in) chemical energy	60%

3.4.2.3 Nuclear hardening. Requirements for nuclear survivability shall be as specified in a separate classified (confidential) document entitled "Nuclear Survivability Criteria for the Automatic Fire Extinguishing System (AFES)" (U) (see 4.7.5.2.3 and 6.6).

3.4.2.4 Slow growth pan-fire. The OFSA shall generate output signal "A" (see table I) on pins C and D when exposed to a slow growth pan-fire. The pan shall be 130 mm (5 in) in diameter, 76 mm (3 in) deep, contain 840 cm^3 (51 in^3) of diesel fuel, and be placed 380 mm (15 in) from the OFSA. The response shall occur after the fire has grown to 50 to 100 percent (%) of full pan size. With the pan at 1200 mm (48 in) from the OFSA, no large fire signal shall be produced on pin C when the fire has grown to full pan size (see 4.7.5.2.4).

3.4.3 Start-up. The OFSA shall not false alarm upon the application of 15 to 30 volts (V) direct current (dc) step inputs across pins A and B (initial power-on condition) (see 4.7.5.3).

3.4.4 Input power. The OFSA shall be effective over an input voltage range of 15 to 30 V dc and shall not require more than 150 milliamperes (mA) when active. The OFSA shall not draw more than 15 mA nor less than 7.5 mA during quiescent operation (see 4.7.5.4).

3.4.4.1 Input voltage decay. The OFSA shall not false alarm (produce small or large fire output signals) when undergoing input voltage decays resulting from prolonged operation without recharging of the battery (see 4.7.5.4.1).

3.4.4.2 Steady state voltage extremes. The OFSA shall sustain no damage from the application of any steady state voltage up to 40 V dc. Subsequently, the OFSA shall meet the requirements of 3.4.1 and 3.4.2 (see 4.7.5.4.2).

3.4.5 Electromagnetic interference (EMI). The OFSA shall meet the following requirements of MIL-STD-461, Notice 4:

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Conducted emission:	CE01, CE04
Conducted susceptibility:	CS01, CS02, CS06
Radiated emission:	RE02, RE02.1
Radiated susceptibility:	RS02, RS03, RS03.1 non-sheltered

In addition, the Narrowband Conducted Emissions shall be in accordance with the limits of figure 4 (see 4.7.5.5).

3.5 Environmental. The OFSA shall produce no false alarms during or after exposure to the following environments. The OFSA shall function properly, evidence no deterioration, and meet the requirements of 3.4.1 and 3.4.2.1 after exposure. The OFSA shall also demonstrate the ability to detect fires and meet the requirements of 3.4.1.3 during exposure to the environments of 3.5.1, 3.5.2, and 3.5.8 (see 4.7.6).

3.5.1 Temperature

3.5.1.1 High temperature. The OFSA shall meet the requirements of 3.5 during and after extended exposure to a temperature of 125°C (257°F) (see 4.7.6.1.1).

3.5.1.2 Low temperature. The OFSA shall meet the requirements of 3.5 during and after extended exposure to a temperature of -51°C (-60°F) (see 4.7.6.1.2).

3.5.1.3 Temperature shock. The OFSA shall meet the requirements of 3.5 when exposed to rapid changes of temperature between -51 and 71°C (-60 and 160°F) (see 4.7.6.1.3).

3.5.2 Vibration. The OFSA shall meet the requirements of 3.5 when exposed to vibration at the amplitudes and frequencies shown in figure 5 in each of three mutually perpendicular axes (see 4.7.6.2).

3.5.3 Shock. The OFSA shall meet the requirements of 3.5 when subjected to sawtooth shock pulses of 100 ± 10 g, with a duration of 11 ± 1.1 ms. The pulses shall be applied in both directions of three mutually perpendicular axes (see 4.7.6.3 and figure 6).

3.5.4 Leakage.

3.5.4.1 Water immersion. The OFSA shall meet the requirements of 3.5 and shall not leak when immersed in water (see 4.7.6.4.1).

3.5.4.2 Diesel fuel immersion. The OFSA shall meet the requirements of 3.5 and shall not leak when immersed in diesel fuel conforming to DF-2 of VV-F-800 (see 4.7.6.4.2).

3.5.4.3 Water jet. The OFSA shall meet the requirements of 3.5 and shall not leak when exposed to water jet cleaning (see 4.7.6.4.3).

3.5.5 Salt fog. The OFSA shall meet the requirements of 3.5 and shall show no evidence of corrosion after prolonged exposure to a salt laden atmosphere (see 4.7.6.5).

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3.5.6 Fungus The OFSA shall meet the requirements of 3.5 following prolonged exposure, in a non-operating mode, to an environment favorable to fungus growth (see 4.7.6.6).

3.5.7 Sand and dust The OFSA shall meet the requirements of 3.5 after exposure to air driven sand and dust. There shall be no damage to the OFSA (see 4.7.6.7).

3.5.8 Humidity. The OFSA shall meet the requirements of 3.5 when operated in a high humidity environment (see 4.7.6.8).

3.6 Finish All component exterior surfaces of the OFSA, except lens windows, mounting surfaces, and electrical connector, shall be painted in conformance with the requirements of Drawing 12370458 (see 4.7.2).

3.7 Identification nameplate and markings

3.7.1 Nameplate. The OFSA shall have a nameplate attached in a conspicuous location. The nameplate shall have the following minimum information (see 4.7.2):

Component Name

National Stock Number (NSN)

Serial Number

Manufacturer's Name or Identification Code

3.7.2 Marking The OFSA shall be marked in conformance with the applicable drawing and MIL-STD-130 (see 4.7.2)

3.8 Workmanship Workmanship shall assure a product free of burrs, scratches, sharp edges, chips, corrosion, scale, and dirt, and shall meet requirement 9 of MIL-STD-454 (see 4.7.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items must meet all requirements of sections 3 and 5. The inspections set forth in this

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specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2. Test equipment and standard conditions. Use of equipment, equipment designs and procedures other than as specified herein shall be subject to approval of the contracting officer.

4.1.2.1 Special test equipment. Special test equipment as well as commercial test equipment is required to perform some of the qualification tests.

4.1.2.2 Electromagnetic shutter and scalloped wheel assembly. This equipment, or equivalent, is required to perform response time tests and is illustrated in figures 7 and 8. Critical to satisfactory performance is the ability to provide a 90° FOV. The time required for the edge of the wheel scallop (see figure 8) to move from a point at the edge of the FOV to the center of the FOV shall not exceed 0.50 ms.

4.1.2.2.1 Equipment for the response characteristics and response time tests. Standard or commercial test apparatus required for the response characteristic (small and large fires) and response time tests are given in table III.

TABLE IV. Components.

Description	Model number and manufacturer
Sixty-two millimeter circular shutter, normally closed; with light emitting diode at the full-open position acting as a synchronization switch	Uniblitz Model 262X2H25X5; Vincents Associates, Rochester, NY or equivalent
Shutter control unit (shutter timing and drive unit	Uniblitz Model SD-1000; Vincents Associates; Rochester, NY or equivalent
Mounting bracket	Model 55529: Electro Corporation Birmingham, MI or equivalent
Solid state - DI - PROX Proximity switch, 12 V dc	Model 55525: Electro Corporation Birmingham, MI or equivalent

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4.1.2.2.2 Starting circuit for the special test equipment. The proximity switch indicated in figures 7, 8, and 9 and table III, used with the scalloped wheel for system response time tests, outputs a 0.1 ms pulse as the edge of the metal triggers the switch (see 6.5.4). The starting circuit is required to process the 0.1 ms pulse and extend the pulse width for a sufficient period of time to enable the shutter to react and come to a full open position.

4.1.2.2.3 Use of special equipment indicated in figure 3. Test equipment specified in 4.1.2.2.2 may be used to perform the tests described in 4.7.5.1 and illustrated in figure 3. These tests are to determine if the system responds to a fire stimuli and the approximate (± 1 s) time frame. These are not response time tests, which are covered in another section of this specification. The starting circuit of 4.1.2.2.2 is not required. The shutter assembly and the OFSA under test shall be mounted to the same base plate, so when the shutter assembly with base plate is rotated about a fixed pivot to obtain the 90° FOV, the OFSA automatically shall rotate the same amount. The OFSA shall then be mounted at right angles to its original position with the optical window facing the same direction (the preceding shall be repeated to obtain the FOV in this second axis). Instrumentation is not required, a stopwatch is sufficient.

4.1.2.2.4 Use of special equipment for response time tests. To determine the response time of the OFSA, as indicated in 4.7.5.1.5 and illustrated in figures 7, 8, and 9, the following test methods or procedures shall be employed using the special equipment and instrumentation of 4.1.2.2, 4.1.2.2.1 and 4.1.2.2.2, or equivalent.

a. Oscilloscope setup procedure. The oscilloscope shall be set at 50 ms full sweep time when the scalloped wheel is run at 1225 to 1750 revolutions per minute (rpm). The oscilloscope shall be triggered by the control unit of the shutter assembly which initially is in the normally closed position, shielding the OFSA from viewing the pan-fire. The oscilloscope shall simultaneously measure and record the original triggering signal, the voltage or current output of the OFSA, the duration of time that the OFSA is exposed to the fire stimuli by the scalloped wheel, and the scalloped wheel rpm. The preceding recordings shall be photographed.

b. Oscillograph setup procedure. The oscillograph shall be programmed identically to 4.1.2.2.4a, along with the time base [1000 ± 5 hertz (Hz)] and the data shall be measured and recorded simultaneously. Recorder paper speed shall be not less than 800 millimeters per second (mm/s) [32 inches per second (in/s)]. The paper recordings shall be duplicated or preserved in such a matter that they will not fade. The combined transient response time constant of all instruments for any channel being recorded shall not be greater than 0.06 ms.

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c. Location and use of equipment indicated in figure 7. The optical shutter with control unit shall be instrumented through the use of a proximity switch and starting circuit, to provide a step input voltage to the solenoid coil of the optical shutter assembly. The proximity switch shall be positioned so that the electric pulse generated by the switch will occur as the trailing edge of the scallop just covers the OFSA. This is accomplished at wheel speeds of 1225 to 1750 rpm. Approximately 20 to 25 ms are required for the shutter leafs to become fully open. This permits the OFSA to be fully exposed by the open shutter; however, it is still shielded from the pan-fire by the unscalloped portion of the wheel (for an additional 2 to 5 ms). Exposure of the OFSA to the pan-fire shall be approximately 20 ms, after which the scallop wheel shall again cover the OFSA.

4.1.2.3 Apparatus. In addition to standard measuring instruments (see table IV) test apparatus shall include chambers for conducting low temperature, high temperature, sand and dust, fungus, humidity, and salt fog tests (see 4.7.6).

4.1.2.4 Calibration of test equipment. Standard instrument tolerance accuracies are listed in table IV. Records of calibration of test instruments at intervals sufficient to establish the required accuracy shall be made available to the Government for review.

TABLE IV Measuring instrument tolerance.

Measuring instrument	Accuracy
Voltmeter ac	± 0.25%
Ammeter ac	± 0.50%
Milliammeter	± 0.50%
Temperature indication system	± 1°C (± 1.8°F)
Timing counter	± 1.0%
Oscilloscope (four or six channel, time measuring)	± 3.0%
Oscillograph (multi-channel)	± 1.0%
Tachometer - generator	± 1.5%
Time measuring system	± 1.0%
Weighing system	± 0.25% gram at 150 to 1500 grams (±0.009 ounce at 5 to 53 ounces)

4.1.3 Qualified products. When a part or component is specified to conform to a specification having a QPL, the contractor shall make available to the Government documentation of item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order and quantity.

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4.2 Classification of inspection:

a. Qualification inspection (see 4.4).
b. Quality conformance inspections (QCI) (see 4.5).
   1. Examination (see 4.5.1)
   2. Tests (see 4.5.2).
c Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a Air temperature: 23 ± 10°C (73 ± 18°F)
b. Barometric pressure: 725 + 50 millimeters of mercury (mm Hg)
- 75
[28.5 + 2 inches of mercury (in Hg)]
- 3
c Relative humidity. 50 ± 30%.

4.3.1 Test temperature stabilization. Tests shall be conducted after the test sample has been allowed sufficient time for temperature stabilization Test temperature stabilization is the point at which the test sample will not vary in temperature more than 2°C (3.6°F) per hour

4 4 Qualification inspection. Qualification inspection shall consist of all the examinations and tests of this specification, as specified in table V (see 6 3)

4.4.1 Qualification samples Unless otherwise specified in the contract or purchase order (see 6.2., four qualification samples, each sample consisting of four OFSAs for a total of 16 OFSA. shall be submitted Each time a test in 4.7 requires testing of an OFSA, all four OFSAs of each sample set shall be subjected to the same test. For type I OFSAs, an additional 12 OFSAs shall be submitted for testing of conformance with 3.4.1.6 and 3 4.2.2. The qualification samples produced with equipment and procedures used in production shall be forwarded to a test facility set forth in the letter of authorization to submit samples. The samples shall be plainly identified by securely attached durable tags marked with the following information:

Samples submitted by (name) (date for qualification inspection in accordance with the requirements of MIL-S-62546(AT) and number under authorization (reference authorizing letter and number).

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TABLE V. Classification of inspections.

Title	Requirement	Inspection	Qualification samples 1 2 3 4	Quality conformance: Examination	Quality conformance: Acceptance Tests	Control
Materials and construction	3.2 thru 3.2.2, 3.3, thru 3.3.2 and 3.3.7	4.7.1	X			
Defects (see table VI)	3.3.1, 3.3.2, 3.6 3.7 and 3.8	4.7.2	X X X X	X		
Features	3.3.1, 3.3.3 3.3.4, 3.3.6	4.7.3	X X			X
Reliability	3.3.5	4.7 4	X X X X			
Performance	3.4	4.7.5				
Response characteristics	3.4.1	4.7.5.1				
FOV	3.4.1.1	4.7.5.1.1	X X X X			X
Thresholds	3.4.1.2	4.7.5.1.2				
Small fire	3.4.1.2.1	4.7.5.1.2.1	X X X X		X	
Large fire	3.4.1.2.2	4.7.5.1.2.2	X X X X		X	
BITE	3.4.1.3	4.7.5.1.3	X X X X		X	
Sensitivity	3.4.1.4	4.7.5.1.4	X X X X			X
Response time	3.4.1.5	4.7.5.1.5	X X X X		X	
Penetration response time	3.4.1.6	4.7.5.2.2.6	X X X X			
False alarm susceptibility	3.4.2	4.7.5.2				
Radiation stimuli	3.4.2.1	4.7.5.2.1	X X X X			X
Penetrating munitions (type I OFSA only)	3.4.2.2	4.7.5.2.2	X X X X			
Nuclear hardening	3.4.2.3	4.7.5.2.3	X X			X
Slow growth pan-fire	3.4.2.4	4.7.5.2.4	X X X X			X
Start-up	3.4.3	4.7.5 3				
Input power	3.4.4	4.7.5.4	X X X X			
Input voltage decay	3.4.4.1	4.7.5.4.1	X X			X
Steady state voltage extremes	3.4.4.2	4.7.5.4.2	X X			X

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TABLE V. Classification of inspections - Continued.

Title	Requirement	Inspection	Qualification samples 1 2 3 4	Quality conformance Examination	Quality conformance Acceptance Tests	Control
EMI	3.4.5	4.7.5.5	X X			
Environmental	3.5	4.7.6				
Temperature	3.5.1	4.7.6.1				
High temperature	3.5.1.1	4.7.6.1.1	X X			X
Low temperature	3.5.1.2	4.7.6.1.2	X X			X
Temperature shock	3.5.1.3	4.7.6.1.3	X X			X
Vibration	3.5.2	4.7.6.2	X X			X
Shock	3.5.3	4.7.6.3	X X			X
Leakage	3.5.4	4.7.6.4				
Water immersion	3.5.4.1	4.7.6.4.1	X X X X		X	
Diesel fuel immersion	3.5.4.2	4.7.6.4.2	X X X X			
Water jet	3.5.4.3	4.7.6.4.3	X X X X			X
Salt fog	3.5.5	4.7.6.5	X X			
Fungus	3.5.6	4.7.6.6	X X			
Sand and dust	3.5.7	4.7.6.7	X X			
Humidity	3.5.8	4.7.6.8	X X			

4.4.2 Qualification inspection failure. Failure of any sample to pass the examination and tests of table V shall be cause for refusal to grant qualification.

4.5 QCI. QCI shall include the examination of 4.5.1 and the acceptance tests of 4.5.2 (see 6.2).

4.5.1 Examination. The OFSA shall be examined for the defects specified in table VI. Noncompliance with any of the requirements specified in sections 3 and 5 shall be cause for rejection.

TABLE VI. Classification of defects

Category	Defects	Method of Inspection
Critical	None	
Major 101	Incorrect dimensions affecting interchangeability (see 3.3.2).	SIE 1/

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TABLE VI. Classification of defects - Continued.

Category	Defects	Method of Inspection
102	Damaged connector (see 3.3.1).	Visual
103	Damaged (cracked, scratched, broken) OFSA window(s) (see 3.3.2)	Visual
104	Improper nameplate (see 3.7.1).	Visual
105	Improper identification markings (see 3.7.2)	Visual
Minor		
201	Incorrect dimensions not affecting interchangeability (see 3.3.2)	SIE
202	Improper finish (see 3.6)	Visual
203	Faulty workmanship (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.2 Acceptance tests Each OFSA shall be subjected to the acceptance tests specified in table V

4.6 Control tests OFSA selected for control tests shall be new and subjected to the control tests specified in table VI (see 6.2)

4.7 Methods of inspection.

4.7.1 Materials and construction Conformance to 3.2 through 3.2.2, 3.3 through 3.3.2 and 3.3.7 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports and rating data

4.7.2 Defects Conformance to 3.3.1, 3.3.2 and 3.6 through 3.8 shall be determined by examination for the defects listed in table VI Examination shall be visual or by measurement with SIE

4.7.3 Features Conformance to 3.3.1, 3.3.3, 3.3.4 and 3.3.6, shall be determined by exercising the OFSA, and by qualitative observation sufficient to demonstrate that specified characteristics and features are present and functional.

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4.7.4 Reliability. To determine conformance to 3.3.5, a reliability prediction analysis shall be conducted in accordance with MIL-HDBK-217 for the OFSA electronic design. A report containing a detailed parts list and reliability calculations for the OFSA shall be available for review by the Government.

4.7.5 Performance

4.7.5.1 Response characteristics. To determine conformance to 3.4.1, the OFSA shall be arranged in the test configuration as indicated in figure 3, and shall meet the requirements of 4.7.5.1.1 through 4.7.5.1.4.

4.7.5.1.1 FOV. To determine conformance to 3.4.1.1, the OFSA shall meet the specified requirements throughout its FOV. The OFSA shall first be tested directly facing the stimulus. The test shall be repeated with the stimulus at ± 45° from straight ahead (in one plane) and then at ± 45° in another plane approximately 90° to the first.

4.7.5.1.2 Thresholds.

4.7.5.1.2.1 Small fire. To determine conformance to 3.4.1.2.1, the OFSA shall output a fire signal at pin D in not more than 5 s in response to a 130-mm (5-in) diameter pan-fire, 76 mm (3 in) minimum depth containing 840 cm^3 (51 in^3) diesel fuel (DF-2 of VV-F-800) at a distance of 1200 mm (48 in) from the OFSA. Pin C shall be monitored for false large fire signals. Time outputs shall be recorded. The FOV shall be checked in accordance with 4.7.5.1.1 by repeating this procedure four times.

4.7.5.1.2.2 Large fire. To determine conformance to 3.4.1.2.2, the procedure of 4.7.5.1.2.1 shall be repeated except that the pan-fire shall be 380 mm (15 in) from the OFSA. The OFSA shall respond by producing Signal "A" at pins C and D as shown in table I.

4.7.5.1.3 BITE. To determine conformance to 3.4.1.3, the OFSA shall be provided with a BITE signal as specified in table I. Voltage, current and duration of signals shall be measured on the small fire and large fire output pins. The test shall be repeated simulating a dirty OFSA window(s). The test shall then be repeated with the optical lens or window facing directly into the sun.

4.7.5.1.4 Sensitivity

4.7.5.1.4.1 Sunlight. To determine conformance to 3.4.1.4a, the tests in 4.7.5.1.1 and 4.7.5.1.2 shall be repeated while the OFSA is facing directly into the sun.

4.7.5.1.4.2 Diesel fuel immersion. To determine conformance to 3.4.1.4b, the tests of 4.7.5.1.1 and 4.7.5.1.2 shall be repeated after the OFSA window(s) has been immersed in diesel fuel (DF-2 of VV-F-800) three times and air dried between each immersion, without cleaning the optical surfaces.

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4.7.5.1.5 Response time. To determine conformance to 3.4.1.5, the OFSA shall be tested as specified in 4.1.2.2.5. The pan-fire specified in 3.4.1.5 shall be placed at a distance of 380 mm (15 in) from the OFSA to the centerline of the pan. The test shall be repeated four times in accordance with 4.7.5.1.1 to check response at the FOV limits. The tests shall be repeated at -17 and 71°C (1 and 160°F) at the FOV limits. The tests shall then be repeated at 15 and 30 V dc at the FOV limits.

4.7.5.2 False alarm susceptibility

4.7.5.2.1 Radiation stimuli. To determine conformance to 3.4.2.1, the threshold immunity distance of the OFSA shall be tested to determine the small fire and the large fire sensitivity limits so that the system can be operated safely without concern of false alarms (see 6.5.3). Unless otherwise specified herein, three types of tests shall be conducted with each non-transient radiation source listed in table II: system on, source turned on and off; source on, system turned on and off (see 4.7.5.2.1.15) and chopped light. The FOV shall be checked in accordance with 4.7.5.1.1 during all tests except for ballistic tests and chopped light tests which shall be conducted only in the straight ahead sensor position. During chopped light tests using the fixture indicated in figure 10, the OFSA shall be exposed for a period of not less than 20 s total exposure time at each increment of light chopper rpm. The chopped light tests shall be conducted as specified in requirement steps 16 and 17 of table II. The OFSA shall not respond at distances equal to or beyond those specified in 3.4.2.1 and table II.

4.7.5.2.1.1 Headlight. Headlight MS53023-1, shall be placed directly in front of the OFSA so that the beam directly hits the OFSA at a distance of 300 mm (12 in). The headlight shall be connected to a 28 V dc power source. The light shall twice be turned on by means of an in-line switch, left on for not less than 5 s, and then turned off for not less than 5 s. The light shall then be turned on and left on, and the OFSA shall be turned on and off twice with 5 s between each switching action. This test shall be repeated four times in accordance with 4.7.5.1.1 to check the 90° FOV of the OFSA. The distance between the OFSA and light source shall be changed to the large fire immunity distance [100 mm (4 in)], and the test shall be repeated. The chopped light test shall be conducted at speeds from near zero to 400 rpm in 50 rpm increments. The small and large fire thresholds shall meet the requirements of step 1 and 16, tale II of 3.4.2.1.

4.7.5.2.1.2 Sun. The OFSA shall be positioned to directly face the sun. The OFSA shall be turned on and off twice with 10 s between each switching action. Then the OFSA shall be switched on and rotated for 20 cycles in a 90° arc within the FOV at a rate of approximately 90° per second. The chopped light simulator tests, straight ahead only, as indicated in 4.7.5.2.1 shall be conducted. The OFSA shall meet the requirements of 3.4.2.1, table II, step 2.

4.7.5.2.1.3 Simulated hatch opening. The procedure in 4.7.5.2.1.9 shall be repeated, except the light sources shall be replaced by direct sunlight simulating an ambient light extreme with vehicle hatch covers open and/or ramp down. The OFSA shall be placed in a horizontal position with the following foreground, at separate times:

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a. Green grass.
b. Desert sand, simulating desert glare.
c. Snow.
d. Fresh (sweet) water.

The OFSA shall meet the requirements of table II, steps 8 and 17.

4.7.5.2.1.4 Bright clothing. The colored materials (nylon cloth) specified below shall be placed, at separate times, directly in front of the OFSA at a distance of 300 mm (12 in).

a. Fluorescent orange: MIL-P-40061, US Army shade 230.
b. Fluorescent red: MIL-P-40061, US Army shade 229.
c. Bright yellow: Commercial grade.

A 40 watt (W) fluorescent light with white enamel reflector, placed 300 mm (12 in) above and 300 mm (12 in) behind the OFSA, shall be beamed at the material in such a manner that the highest reflected light intensity shall be seen by the OFSA. The fluorescent light shall be switched on and off for 5 s each, 10 times. The light shall then be turned on and remain on while the OFSA is turned on and off 10 times with 5 s between each switching action. The OFSA or material will be moved inward in increments of 50 mm (2 in) until 50 mm (2 in) distance is reached. The test shall be repeated at each increment including the 50 mm (2 in) distance. The chopped light simulator test indicated in 4.7.5.2.1 shall be conducted at each increment from 300 to 50 mm (12 to 2 in) distance while the light is turned on. The entire test shall then be repeated with the following combination of light sources replacing the fluorescent light: Headlight (MS53023-1) and flashlight (MX 991/U), headlight and incandescent light (frosted, 100 W); headlight and movie light (625 W DWY quartz); and a movie light and incandescent light. The OFSA shall meet the requirements of table II, step 9.

4.7.5.2.1.5 Colored material. The colored material of 4.7.5.2.1.4 shall be placed in front of the OFSA in the same manner as the headlight in 4.7.5.2.1.1, except the light source shall be bright sunlight. The OFSA shall be placed so that the sun is above and behind the OFSA at 45 to 60° from the horizontal. The OFSA shall be moved from side to side and rotated in view of the material and then out of view, as rapidly as possible for 30 s. The 90° FOV shall be checked in accordance with 4.7.5.1.1. The chopped light simulator test as indicated in 4.7.5.2.1.1 shall be repeated. The OFSA shall meet the requirements of 3.4.2.1, table II, step 9.

4.7.5.2.1.6 Flashlight. The flashlight shall be in accordance with MIL-F-3747, type I, style 2-2 cell (MX 991/U) and the batteries shall be in accordance with MIL-B-18/9 (BA-30). The flashlight, using the red lens over the clear lens, shall be positioned directly in front of the OFSA so that the beam hits the OFSA directly at near zero distance. The flashlight shall twice be turned on for not less than 5 s and then turned off. The flashlight shall then be turned on and remain on while the OFSA is turned on and off twice with 5 s between switching actions. This test shall be repeated to check the FOV limits in accordance with 4.7.5.1.1. A test shall be conducted

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with the flashlight in the on position continuously using the chopped light simulator as indicated in 4.7.5.2.1. The requirements of step 14 and 16, table II shall be met.

4.7.5.2.1.7 Reflected light. A white enamel reflector, measuring 600 by 600 mm (24 by 24 in) square, shall be placed vertically in front of the OFSA at a distance of 600 mm (24 in). The following light sources, simultaneously, shall be placed 300 mm (12 in) above and 300 mm (12 in) behind the OFSA in an area not greater than 1200 by 1200 mm (48 by 48 in):

a. Vehicle headlight, MS53023-1.
b. Vehicle infrared (IR) light, MS53024-1.
c. Incandescent light, frosted glass, standard household, 100 W.
d. Incandescent light, clear glass, 100 W.
e. Fluorescent light, 40 W.
f. Electronic flash (intermittently).
g. Movie light.

The light sources shall be switched on for 5 s and off for 5 s, 10 times. The lights shall then be turned on and remain on while the system is turned on and off 10 times with 5 s between each switching action. The tests shall be repeated to check the 90° FOV in accordance with 4.7.5.1.1. The light sources shall be moved in to a distance of 150 mm (6 in) from the reflector and the test repeated. After the last sequence, the chopped light simulator test, as indicated in 4.7.5.2.1, shall be conducted using the light sources indicated in this paragraph. The OFSA shall meet the requirements of table II, steps 15 and 16.

4.7.5.2.1.8 Miscellaneous light sources. The light sources specified in table VII shall be separately placed directly in front and facing the OFSA at the small fire and then the large fire immunity distances, as indicated in 3.4.2.1 and table II, and switched on for 5 s and then off for 5 s, 10 times. The source, if applicable (see 4.7.5.2.1.15), shall then be turned on and left on while the OFSA is turned on and off twice with 5 s between switching actions. The OFSA FOV shall be checked in accordance with 4.7.5.1.1 at both distances. The light sources while in the on-position (except for the electronic flash which shall be operated intermittently), shall be exposed to the chopped light simulator as indicated in 4.7.5.2.1.1. The OFSA shall meet the requirements of 3.4.2.1 and table II.

4.7.5.2.1.9 Incandescent and IR light sources. The following light sources, simultaneously, shall be placed at a distance of 600 mm (24 in), in an area that is not beyond the 90° FOV of the OFSA:

a. Vehicle IR light, MS53024-1.
b. Vehicle headlight, MS53023-1.
c. Incandescent light, frosted glass, standard household, 100 W.
d. Fluorescent light, standard office or shop with white enamel reflector, 40 W.
e. Incandescent light, clear glass, 100 W.

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The OFSA shall be shielded from the light sources. The shield shall be removed as rapidly as possible by hand, leaving the OFSA exposed to the sources for 1 s and then the shield shall be replaced in the same manner. This procedure shall be repeated nine times The shield shall next be removed and the OFSA shall be turned on and off 10 times with 5 s between each switching action Then the lights shall be left on and the chopped light simulator test in 4.7.5.2.1 shall be conducted, using the light sources listed above The OFSA shall meet the requirements of table II, steps 8 and 17.

TABLE VII. Lighting.

Test step	Requirement 3.4 2.1 step	Description	Distance [mm(in)]	
			Small fire	Large fire
1	3, 16	Incandescent frosted glass light, standard household, 100 W	150 (6)	25 (1)
2	4, 16	Incandescent clear glass light, 100 W	225 (9)	50 (2)
3	5, 16	Fluorescent light with white enamel reflector, standard office or shop, 40 W (or two 20 W)	150 (6)	IAD
4	7, 16	Vehicle IR light conforming to MS53024-1	600 (24)	300 (12)
5	12, 16	Red dome light conforming to MS51073-1	IAD	IAD
6	13, 16	Blue-green dome light conforming to MS51073-1	IAD	IAD
7	6, 16	Electric arc [12 mm (15/32 in) gap at 4000 V alternating current]	25 (1)	25 (1)
8	10, 16	Electronic flash (180 watt-seconds minimum output)	450 (18)	225 (9)
9	11, 16	Movie light, 625 W quartz DWY lamp, (Sylvania S.G.-55 or equivalent)	1200 (48)	600 (24)

4.7.5.2.1.10 Heaters. A 1500-W radiation heater shall be placed in front of the OFSA at a distance of 900 mm (36 in) for the small fire and 450 mm (18 in) for the large fire immunity distance. The 90° FOV shall be

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checked in accordance with 4.7.5.1.1. At each position, orientation and distance, the OFSA shall be turned on and off twice with 10 s between each switching action. The test shall be repeated with a 1000-W radiation heater with a fan placed in front of the heater blowing towards the OFSA, with the distance of 600 mm (24 in) for the small fire and 300 mm (12 in) for the large fire threshold. The OFSA shall meet the requirements of table II, steps 18 and 19.

4.7.5.2.1.11 Welding. Two of the following, including (a), shall be welded by both the arc and acetylene methods in a vertical or horizontal lap-weld forming a 12 mm (15/32 in) bead, 300 mm (12 in) long. The material shall be centered directly in front of the OFSA at distances specified in table II, steps 20 and 21. The test shall be repeated while the OFSA is turned on and off with 5 s between switching actions. The OFSA shall meet the requirements of table II, steps 20 and 21.

a. Two armor steel plates, 4130 steel, measuring 300 by 100 by 25 mm (12 by 4 by 1 in), welded with Number 307 or 308 stainless steel rod having a 4 mm (5/32 in) diameter and using 300 A [also 00 tip, 16 by 150 mm (5/8 by 6 in) flame].

b. Two armor aluminum plates, 5083 measuring 300 by 100 by 25 mm (12 by 4 by 1 in), welded with Number 5356 rod having a diameter of 2.4 mm (3/32 in) [00 tip, 16 by 150 mm (5/8 by 6 in) flame].

c. Two armor aluminum plates, 7039 measuring 300 by 100 by 25 mm (12 by 4 by 1 in), welded with Number 5356 rod having a diameter of 2.4 mm (3/32 in).

4.7.5.2.1.12 Ballistics.

a. An M16 rifle shall be placed in a horizontal position with the OFSA located 250 mm (10 in) from the end of the barrel and facing the rifle flash along a line 53 ± 5° from the path of the projectile. While in this position, the M16 rifle shall be fired at least 10 times for the small fire threshold. For the large fire threshold, the OFSA shall be 50 mm (2 in) beside and facing directly at the M16 rifle. The M16 rifle shall be fired at least 10 times. The OFSA shall meet the requirements of table II, step 22.

b. The OFSA shall be exposed to a 105-mm (4-in) muzzle flash 6000 mm (236 in) from the OFSA while the OFSA is facing the point (area) of muzzle flash. The OFSA shall then be reoriented by 40 to 45° such that the muzzle flash is seen by the edge of the required FOV. The OFSA shall again be exposed to the 150-mm (6-in) muzzle flash at 6000 mm (236 in) distance. If either test produces an OFSA large fire signal, the tests shall be repeated at 7000 mm (276 in). Otherwise, the tests shall be repeated at 5000 mm (197 in). The OFSA shall be in compliance with table II, step 23 if the distances and results of the tests (whether an OFSA large or small fire signal was produced) are recorded.

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c. The OFSA shall be exposed to a 50-caliber muzzle flash at 1000 mm (40 in). Five, four-shot rapid fire bursts shall be fired while covering the 90° FOV as specified in 4.7.5.1.1. The tests shall be repeated at 600 mm (24 in) and at 1400 mm (56 in). The OFSA shall be in compliance with table II, step 24. The distances and results of the tests (whether an OFSA large or small fire signal was produced) shall be recorded.

4.7.5.2.1.13 Cigarette and cigar. A lit cigarette shall be placed directly in front of the OFSA at a distance of 100 mm (4 in) and then at 25 mm (1 in) and the OFSA shall be moved from side to side in approximately a 150 mm (6 in) arc as rapidly as possible for 10 s. Then the OFSA shall be turned on and off twice with 5 s between each switching action while the OFSA continues to be moved in the same manner. This procedure shall be repeated with a lit cigar. The OFSA shall meet the requirements of table II, steps 25 and 26.

4.7.5.2.1.14 Matches. A paperbook match shall be lit at a distance of 300 mm (12 in) and then at 100 mm (4 in) directly in front of, exposing the match flare-up to the OFSA. After lighting, the OFSA shall be moved from side to side as rapidly as possible, without the match going out, for at least 5 s. This procedure shall be repeated with a wooden match. The OFSA shall meet the requirements of table II, steps 27 and 28

4.7.5.2.1.15 Cycle. Tests where the source is kept on while the OFSA is cycled on and off shall be conducted as required in the preceding paragraphs. An exception, however, is made for source indicated in steps 10, 16, 17, 22, 23, 24, 27, and 28 of table II. Where these excepted sources are cited in paragraphs requiring this test (for example, combinations of sources which include an excepted source), those particular sources shall be deleted while the source-on, OFSA cycled on-and-off tests are performed

4.7.5.2.2 Penetrating munitions. To determine conformance to 3.4.1.6 and 3.4.2.2, performance of the OFSA under actual or simulated installation conditions shall be verified by the tests specified below, during and subsequent to armor penetration.

4.7.5.2.2.1 Target enclosure. A reusable target enclosure shall be designed and fabricated for the accomplishment of these tests Government approval shall be obtained for the enclosure design The enclosure shall be rectangular in shape and have an interior volume of 5.7 ± 0.6 cubic meters (m^3) [200 ± 20 cubic feet (ft^3)]. It shall contain a square projectile entrance opening in one side and a square exit opening directly opposite the entrance opening It shall have provisions for attaching removable entrance armor plates on the outside of the enclosure and exit armor plates on the inside. The enclosure shall incorporate a removable 38 liter (10 gallon) minimum capacity fuel tank. The tank shall be located between the penetration entrance and exit plates, at a distance of not less than 1000 mm (40 in) and not more than 1500 mm (60 in) from the point of munition penetration. The OFSAs shall be mounted with a full view of the munition path. The OFSAs shall be no closer than 380 mm (15 in) to the fuel tank, and no further than 1800 mm (71 in) from the point of munition penetration.

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4.7.5.2.2.2 Armor. The aluminum target plates, entrance and exit, shall be 32 mm (1.25 in) thick of either 5083 or 7039 material. The steel target plates shall be 51 mm (2 in) thick rolled homogenous armor (RHA). The steel/aluminum applique entrance and exit plates shall consist of 32 mm (1.25 in) thick RHA over 25 mm (1 in) thick 5083 or 7039 aluminum, an air gap of 90 mm (3.5 in), and then a 19 mm (0 75 in) thick Kevlar spall liner.

4.7.5.2.2.3 Discrimination (Type I OFSAs only) A maximum number of OFSAs shall be tested simultaneously by subjecting them to the armor/threat penetrations specified in table III For each test, large and small fire output signals shall be recorded for each OFSA. The tests shall be repeated until a minimum of 20 OFSA large fire data points (response or no response) are collected for each armor/threat profile. After each shot, inoperable OFSAs shall be replaced. Data from damaged OFSAs shall not be scored.

4.7.5.2.2 4 Penetration response time A maximum number of OFSAs shall be tested simultaneously by subjecting them to the stimuli specified below An 89 mm (3.5 in) chemical energy munition shall be shot through each of the armor configurations described in 4 7 5.2.2.2 and then through a fuel tank containing diesel fuel (DF-2 per VV-F-800) preheated to sufficient volatility to assure a fire For each test, the time from start of munition penetration to ignition of the fire, to OFSA large and small fire output signals for each OFSA shall be recorded After each test, inoperative OFSAs shall be replaced The tests shall be repeated with a 105 mm (4 in) chemical energy round and preheated hydraulic fluid conforming to MIL-H-46170.

4.7.5 2.3 Nuclear hardening (see 6.6).

4.7.5.2.3.1 Qualification. Conformance to 3.4.2.3 shall be determined on the basis of circuit analysis or component testing.

4.7 5.2 3.2 Control Conformance to 3 4.2.3 shall be determined by inspection of contractor records providing certification that the OFSA has been fabricated in accordance with the design verified and approved for qualification

4.7.5 2.4 Slow growth pan-fire Conformance to 3.4.2 4 shall be determined according to 4.7 5.2.4.1 and 4 7.5.2 4 2.

4 7.5.2 4.1 Slow growth fire With 840 cm^3 (51 in^3) of diesel fuel conforming to DF-2 of VV-F-800 at room temperature, in a 130-mm (5-in) diameter pan, 76 mm (3 in) deep, placed 380 mm (15 in) to the centerline of the pan from the OFSA, a fire shall be lit that results in a slow growth rate The air flow shall not exceed 300 mm/s (12 in/s) The OFSA shall not generate a large fire signal until the flame engulfs at least 50% of the full pan diameter.

4 7.5.2 4.2 Extended distance The preceding test procedures shall be repeated except the distance from the OFSA to the centerline of the pan shall be 1200 mm (48 in). The small fire warning shall respond, and no large fire signals shall be generated.

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4.7.5.3 Start-up. To determine conformance to 3.4.3, the OFSA shall stand in an inactive mode for a minimum of 5 minutes. Power shall be applied to pins A and B according to table I. There shall be no output signal at pins C or D. The test shall be conducted at 15, 20, 24, 28, and 30 V dc.

4.7.5.4 Input power. To determine conformance to 3.4.4, the OFSA shall be connected to a variable power supply set at 24 V dc. Voltage and current draw shall be recorded during quiescent operation, during BITE, and during exposure of the OFSA to a large fire stimulus. The test shall be repeated at 15, 28, and 30 V dc.

4.7.5.4.1 Input voltage decay. To determine conformance to 3.4.4.1, the OFSA shall be connected to a variable power supply. With the OFSA in the quiescent mode, set the power supply at 30 V dc. Gradually decrease the voltage to zero over five to ten minutes. There shall be no large or small fire output signals. The test shall be repeated three times.

4.7.5.4.2 Steady state voltage extremes. To determine conformance to 3.4.4.2, the OFSA shall be connected to a variable power supply. The voltage shall be set at 5 V dc for 3 minutes of OFSA quiescent operation. It shall then be increased in 5.0 V dc increments for 3-minute intervals until reaching 40 V dc. The OFSA shall subsequently pass the tests of 4.7.5.1.2, 4.7.5.1.3. 4.7.5.1 5 (at ambient temperature), and 4 7.5.2.1.8 (using light sources 1, 2, 8, and 9 of table VII).

4.7.5.5 EMI To determine conformance to 3.4.5, the OFSA shall be subjected to the test procedures specified in MIL-STD-462, Notice 3, and figure 4. Narrowband conducted emissions of figure 4 shall be verified in accordance with the basic methodology of MIL-STD-462, Notice 3, method CE07. The OFSA shall not false alarm or cause any EMI problem for other vehicle components.

4 7.6 Environmental To determine conformance to 3.5 the OSFA shall be tested as specified below. The OFSA shall be operated in the quiescent mode, except as otherwise specified herein. The OFSA shall demonstrate the ability to detect fires and pass the test of 4.7.5.1.3 during exposure to the environments of 4.7.6.1, 4.7.6.2, and 4.7.6.8. Unless otherwise specified herein, the OFSA shall subsequently pass the tests of 4.7.5.1.2, 4.7.5.1.3, 4.7.5.1.5 (at ambient temperature), and 4.7.5.2.1.8 (using light sources 1, 2, 8, and 9 of table VII).

4.7.6.1 Temperature.

4.7.6.1.1 High temperature. To determine conformance to 3.5.1.1, the OFSA shall be tested in an oven maintained at 125°C (257°F). The OFSA shall have 24 V dc power connected across pins A and B. The oven shall be maintained at high temperature for a period of 200 ± 24 hours. There shall be no false alarms. Record voltage across pins C and D continuously during the test for indications of any false alarms. The sample shall be inspected at least two times per working day, at least 4 hours apart, and the inspection shall be noted on the recorder trace. During these inspections, the OFSA shall be activated by exposing it to an appropriate radiation source, and inputting a BITE activation signal on pin E.

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4.7.6.1.2 Low temperature. To determine conformance to 3.5.1.2, the OFSA shall be placed in a low temperature chamber. The test conditions and instrumentation requirements of the preceding paragraph shall apply except that the temperature shall be lowered and maintained at -51°C (-60°F).

4 7.6.1.3 Temperature shock. To determine conformance to 3.5.1.3, the OFSA shall be put through three high temperature to low temperature cycles as described below. The instrumentation requirements shall be identical to the preceding paragraph as to type(s) and location(s). The OFSA shall be set up and checked with a radiation source which shall activate the large and small fire signals and by inputting a BITE activation signal on pin E. The OFSA shall be exposed to ambient temperature for no more than 5 minutes during transfer. The procedures of 4.7.6.1.1 and 4.7.6.1.2 apply except:

Step 1. The OFSA and appropriate electrical harnesses shall be placed in an oven preheated to 71°C (160°F) and maintained for 24 ± 2 hours While at this high temperature the OFSA shall be tested as described above.

Step 2. Immediately following step 1 the test sample shall be transferred to a precooled low temperature chamber and maintained at -51°C (-60°F) for a period of 24 ± 2 hours While at low temperature the OFSA shall be tested as described above

Step 3. Repeat step 1

Step 4. Repeat step 2.

Step 5 Repeat step 1.

Step 6 Repeat step 2. Fifteen minutes after transfer, the false alarm tests of 4.7.5.2.1.8, using light sources 1, 2, 8, and 9 of table VII shall be conducted. At the end of Step 6 the OFSA shall be returned to ambient temperature and meet the requirements of 4.7.6.

4.7.6.2 Vibration. To determine conformance to 3.5.2, the OFSA shall be mounted in a production installation configuration or equivalent to a vibration exciter capable of producing and transmitting the specified vibrations The OFSA shall be connected as in use and also connected to the instrumentation necessary for this test The vibration acceleration levels shall be in accordance with figure 5, applied in 15 minute sweep times in accordance with figure 11. Resonant search and dwell of 30 minutes at each of the four most severe resonant frequencies shall be part of the total cycle time of 3 hours. The OFSA shall pass the tests specified in 4.7.6 during and after exposure.

4.7.6.3 Shock To determine conformance to 3.5.3, the OFSA shall be exposed to the shock pulse defined in figure 6 three times in each direction of three mutually perpendicular axes. The OFSA shall pass the tests specified in 4.7.6 during exposure.

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4.7.6.4 Leakage. The OFSA shall not be operated during the following tests. The OFSA shall not evidence any damage or leakage. Subsequent to each of these tests, the OFSA shall pass the tests specified in 4.7.6.

4.7.6.4.1 Water immersion. To determine conformance to 3.5.4.1, the OFSA shall be submerged in water by one of the two methods below. Bubbles coming from within the OFSA shall be considered leakage. However, bubbles which result from trapped air on external surfaces of the OFSA shall not be considered leakage.

a. The temperature of the water and the OFSA shall be 23 ± 10°C (73 ± 18°F). The OFSA shall be immersed in water so that the uppermost part of the OFSA is 50 ± 25 mm (2 ± 1 in) below the surface of the water. The initial air pressure above the water shall be reduced to 480 mm (19.0 in) Hg absolute and maintained for 1 minute or until bubbles substantially cease to be given off by the water, whichever is longer. The air pressure above the water shall then be increased to 510 mm (20.0 in) Hg absolute and maintained for 60 minutes.

b. The OFSA shall be tested in accordance with MIL-STD-810, method 512.2, procedure I. The OFSA shall be heated 27°C above the water temperature of 23 ± 10°C (73 ± 18°F) and submerged to a depth of 1 meter (39.4 in).

4.7.6.4.2 Diesel fuel immersion. Prior to immersion in diesel fuel (DF-2 per VV-F-800), the temperature of the OFSA shall be 45 ± 3°C (113 ± 5°F), the temperature of the diesel fuel shall be 18 ± 5°C (64 ± 9°F). The container shall be of sufficient capacity so that immersion of the OFSA shall not raise the fuel temperature more than 3°C (5°F). The uppermost point of the OFSA shall be a minimum of 50 mm (2 in) below the surface. The OFSA shall remain immersed for 120 ± 5 minutes. Upon completion, remove the OFSA from the fuel and wipe the exterior surfaces only. The OFSA shall be weighed before and after the test. The weight shall not vary more than 0.5 gram (0.02 ounce).

4.7.6.4.3 Water jet. To determine conformance to 3.5.4.1, the OFSA shall be subjected to a water jet spray applied at right angles to the surfaces of the non-operating OFSA at a distance of not less than 300 mm (12 in) from the jet. The surface shall be cleaned at not less than 0.02 square meters per minute (0.2 square feet per minute). The water jet shall be derived from a nozzle having an orifice diameter of not more than 6 mm (0.25 in) and a nozzle pressure of 345 ± 105 kPa (50 ± 15 psi). The OFSA shall not leak and the optical window(s) shall not be damaged. Following the test, the OFSA shall be wiped dry. The OFSA shall be weighed before and after this test. The weight shall not vary more than 0.5 gram (0.02 ounce).

4.7.6.5 Salt fog. To determine conformance to 3.5.5, a non-operating OFSA shall be subjected to the test of MIL-STD-810, method 509.2, except the test shall last 200 hours. The OFSA shall subsequently pass the tests specified in 4.7.6.

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4.7.6.6 Fungus. To determine conformance to 3.5.6, a non-operating OFSA shall be subjected to the fungal incubation specified in MIL-F-13927, for a period of 90 days. Subsequently, the OFSA shall show no evidence of biological growth that could result in the operation of the OFSA falling outside of the performance requirements specified herein. The OFSA shall subsequently pass the tests specified in 4.7.6 without cleaning

4.7.6.7 Sand and dust To determine conformance to 3.5.7, an OFSA with connector attached shall be tested in accordance with MIL-STD-810, method 510.2, procedure I. During this test, the OFSA shall be rotated so that the optical window(s) and the connector receive equal blast time. The high temperature portion of testing shall be conducted at 71°C (160°F) The OFSA shall subsequently pass the tests specified in 4.7.6.

4.7.6.8 Humidity. To determine conformance to 3.5.8, the OFSA shall pass the tests specified in 4.7.6 during and after exposure to warm high relative humidity in accordance with the applicable conditioning procedure for ground electronic equipment of MIL-STD-810, method 507. After conditioning, the OFSA shall be exposed to five continuous 48-hour cycles in accordance with figure 12. Prior to post exposure operation, the OFSA shall be conditioned for 24 hours at 23 ± 3 °C (73 ± 5°F) and 50 ± 10% relative humidity.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for the desired level shall be in accordance with the applicable packaging requirement specified by the contracting authority (see 6.2).

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use The OFSA is the sensing element of Halon 1301 fire extinguishing systems used in military ground combat and tactical vehicles [such as the M992 Field Artillery Ammunition Support Vehicle (FAASV)]. The OFSA responds to optical radiation from exploding atomized or vaporized hydrocarbons and energizes the fire extinguishing system within 3 to 4 ms, depending on the OFSA type (see 3.4.1.5), when an energy level equal to or greater than the large fire threshold is attained. The systems are intended for use in personnel areas and other compartments such as engine spaces. These systems are designed to protect against the potential danger of detonations, deflagration, and slow growth fires due to the presence of highly combustible fuels or other liquid hydrocarbons or flammable debris.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. OFSA type (see 1.2).

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c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Responsibility for inspection, if other than as specified (see 4.1).
e. Inspection conditions, if other than as specified (see 4.3).
f. Control test sample size, frequency of tests, and disposition of lots and defective OFSAs (see 4.4.1).
g. Sample size for QCI examination and acceptance tests and QCI acceptance criteria (see 4.5)
h. Selection of applicable level and applicable packaging requirements (see 5.1).

6.3 Qualification With respect to products requiring qualification, awards shall be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable QPL (QPL-62546), whether or not such products have actually been so listed by that date The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered in this specification. The activity responsible for this QPL is US Army Tank-Automotive Command, AMSTA-GDS, Warren, MI 48397-5000 (see 3.1).

6.3.1 Retention of qualification. The retention of qualification shall consist of periodic verification to determine compliance of the qualified OFSA with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the QPL and shall be included in the Notice of Qualification letter

6.4 Supersession data This military specification supersedes MIL-S-62546(AT), 5 April 1988 and that portion of purchase description ATPD-2070, Edition 6, revision A, 29 February 1984: Sensor, Fire, Optical System with Amplifier, Standard Control, Electronic, which pertains to the OFSA.

6.5 Definitions

6.5.1 Discriminatory OFSA (type I). A discriminatory OFSA is described as one that meets the following:

a All requirements of Section 3.
b. THE OFSA shall not respond (see 3.4.2.2) to the penetration and flash of a threat munition that passes into or through steel or aluminum armored vehicles without initiating a hydrocarbon fire.

6.5.2 Non-discriminatory OFSA (type II). A nondiscriminatory OFSA is described as one that shall meet the requirements of Section 3 except for 3.4.2.2.

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6.5.3 False alarm. A false alarm exists when either an OFSA large or small fire signal is produced without proper input stimuli.

6.5.4 Proximity switch. The units used here are solid state proximity switches which are integral units containing both sensing and switching elements The unit senses and measures any ferrous metal object at distances as given by the manufacturer. These are used in several places in the test circuits Units respond at a rate of 40,000 pulses per second with output rise and fall times of 0 005 ms, the output current is not less than 100 mA (see table III)

6 5.5 Recovered materials. 'Recovered materials' means materials that have been collected or recovered from solid waste (see 6.5.6).

6.5.6 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C 1342 et seq), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6 5 7 Pressure All pressures specified in this document are gage, unless otherwise specified.

6.6 Nuclear survivability criteria. Vendors requesting the nuclear survivability criteria, which is classified confidential, shall possess the proper security clearance and a demonstrated need to know (see 2.1.2 and 3.4.2.3)

6 7 Inspection The following information is presented as guidance for the contracting officer in specifying sampling and acceptance procedures for Quality Conformance Examination and Control Testing in acquisition documents.

6 7.1 Quality Conformance examination

6.7.1.1 Lot formation. A lot should consist of all the OFSAs of one type and part or identifying number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

6.7.1.1.2 Sampling for examination. When sampling for examination is required, consideration should be given to selecting samples in accordance with inspection level II of MIL-STD-105

6.7.1.2 Acceptable quality level (AQL) Samples selected in accordance with 6.7.1.1.2 should be examined for the defects listed in table VI. Lot acceptance/rejection shall be based on the following AQLs:

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Classification	AQL
Major	1.0
Minor	2.5

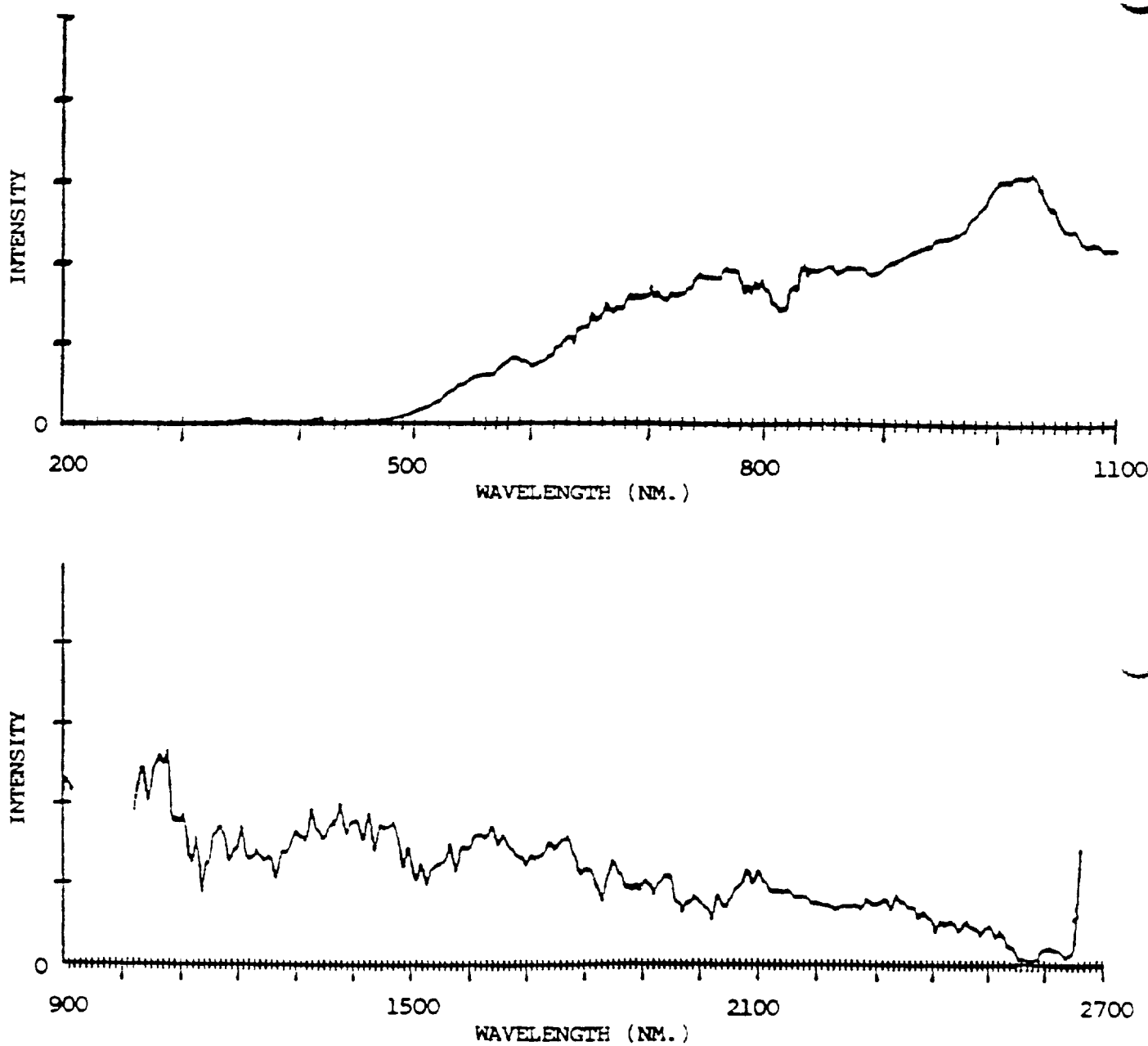
6.7.2 Control tests. For control test purposes, consideration should be given to randomly select one OFSA from each lot of 250 units consecutively produced, except that not more than one OFSA should be tested in a 30-day period, nor less than one OFSA tested in a 6 month period.

6.8 Subject term (key word) listing

Discriminatory/non-discriminatory optical fire sensor
Fire extinguishing system optical sensor
Optical sensor, fire extinguishing system

6.9 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications

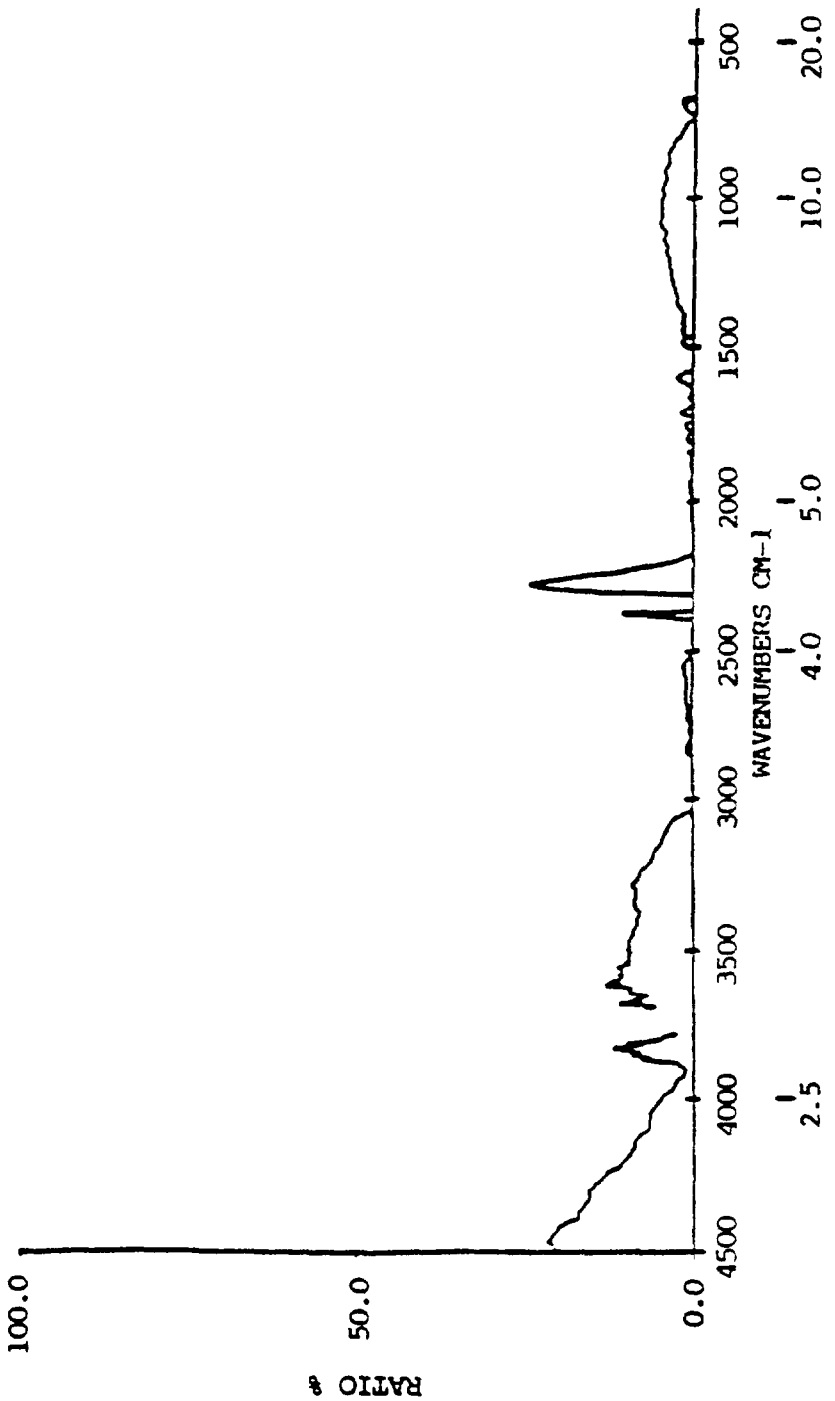
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(The plot is in W/cm^2-sr-nm.)

FIGURE 1. Spectra of diesel fuel, VV-800 DF-2, in the UV, visible, and near IR.

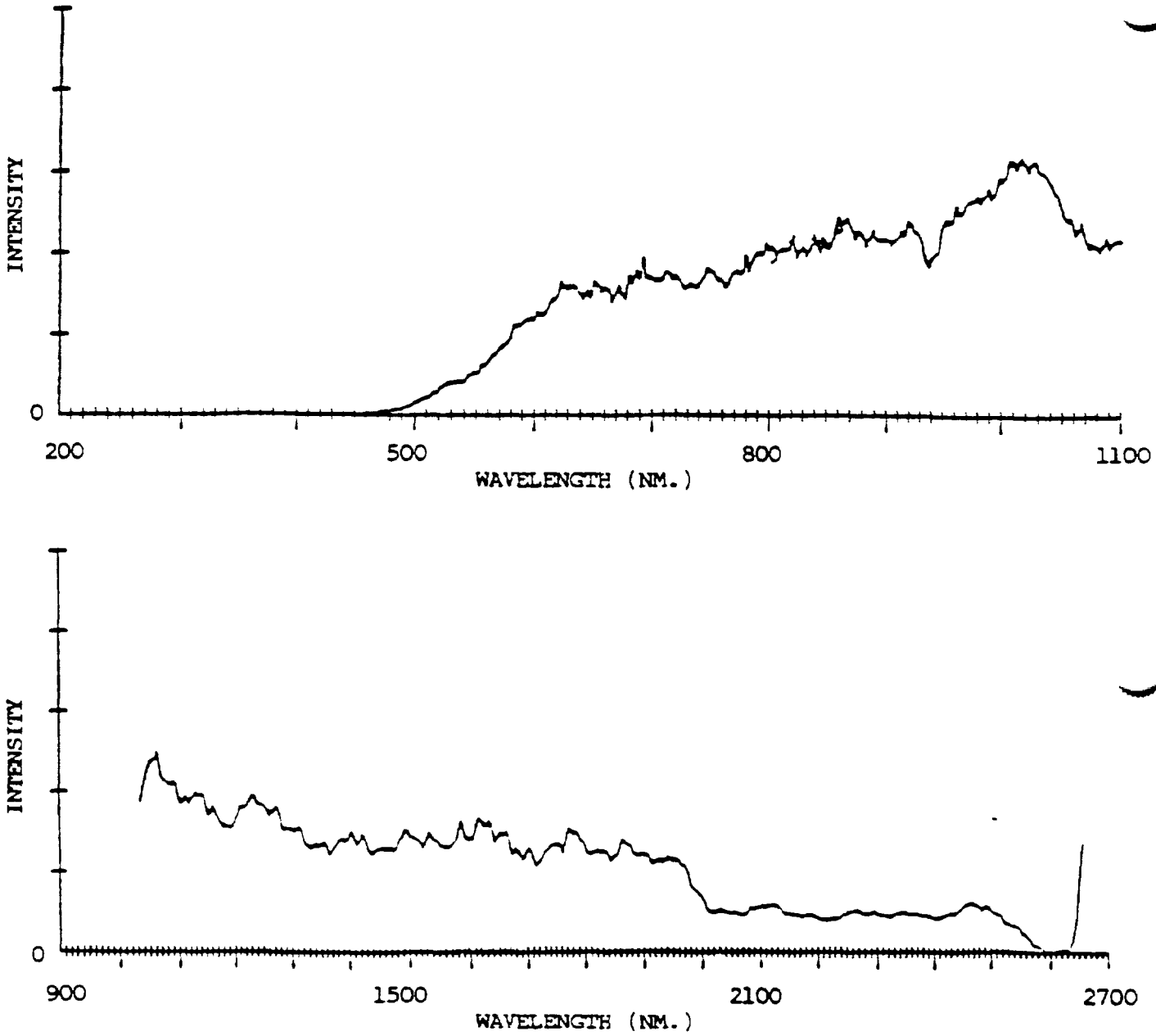
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(FT-IR spectra against a blackbody source; ratio: percent vs wavelength.)

FIGURE 1. Spectra of diesel fuel, VV-800 DF-2, in the infrared region from 2.7 to 20 um. - Continued.

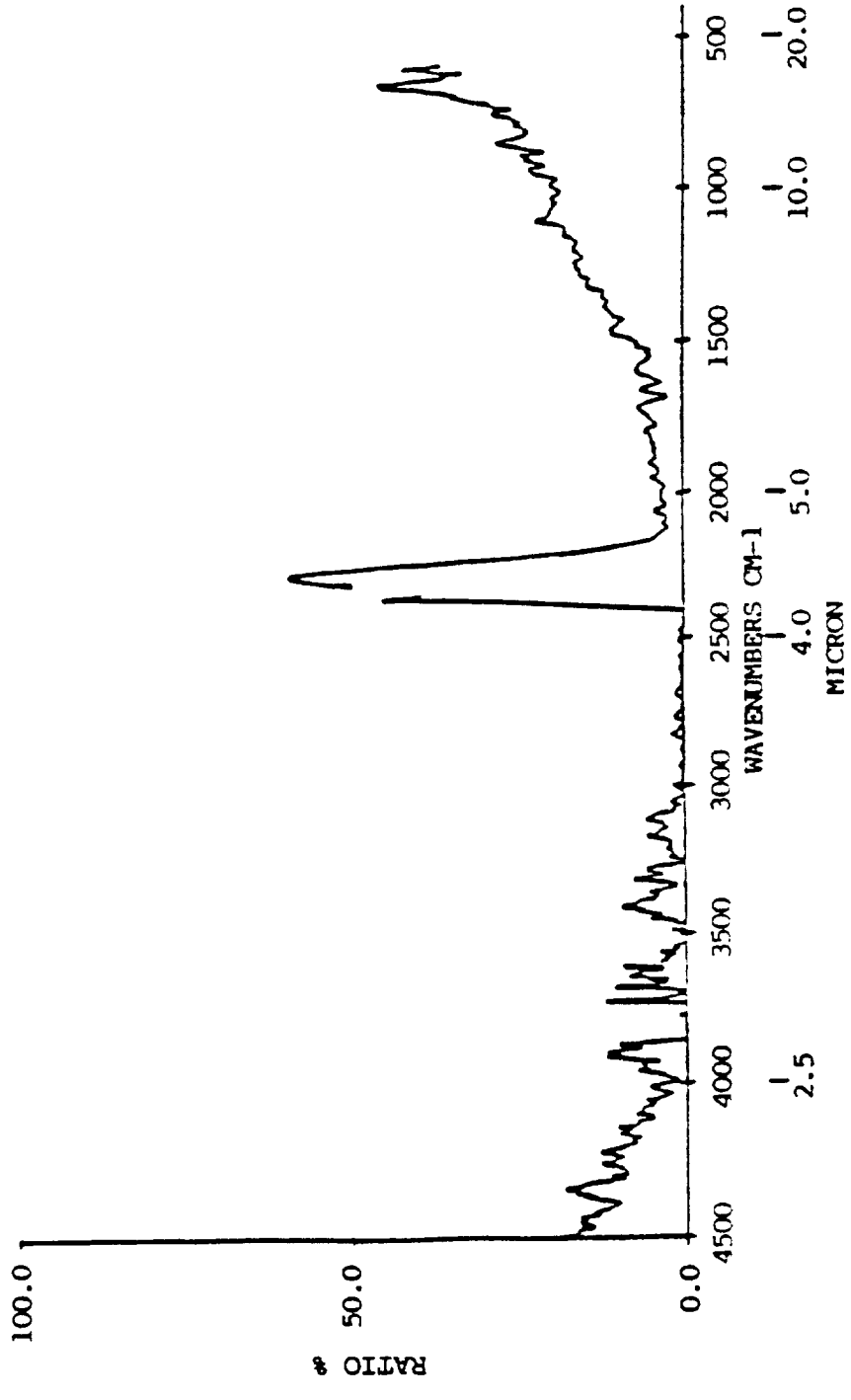
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(Plot is in W/cm^2-sr-nm vs wavelength, nm.)

FIGURE 2. Spectra of hydraulic fluid, MIL-H-5606-E, in the UV, visible, and near infrared.

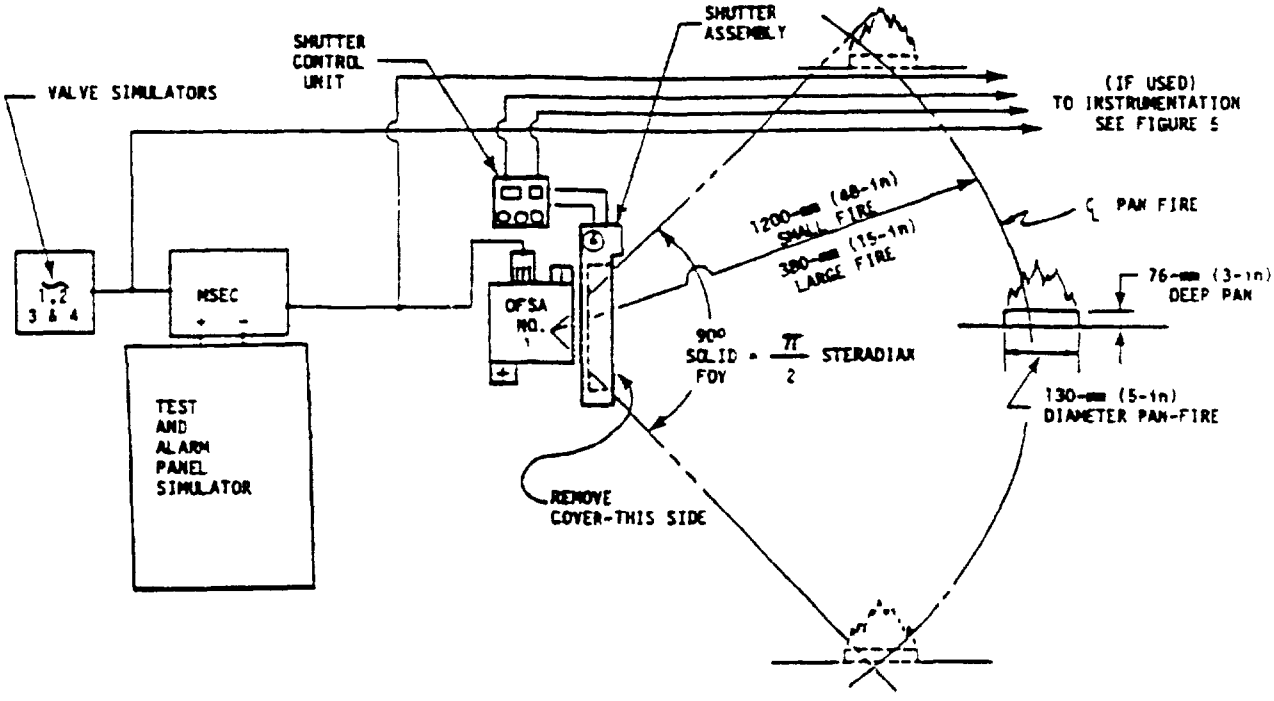
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(FT-IR data ratioed against the globar (blackbody source):
(plot is ratio: percent vs wavelength.)

FIGURE 2. Spectra of hydraulic fluid, MIL-H-5605-E, in the infrared 2.7-20 um. - Continued.

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NOTE: Use of shutter assembly and shutter control unit is optional.

FIGURE 3. Small fire and large fire threshold.

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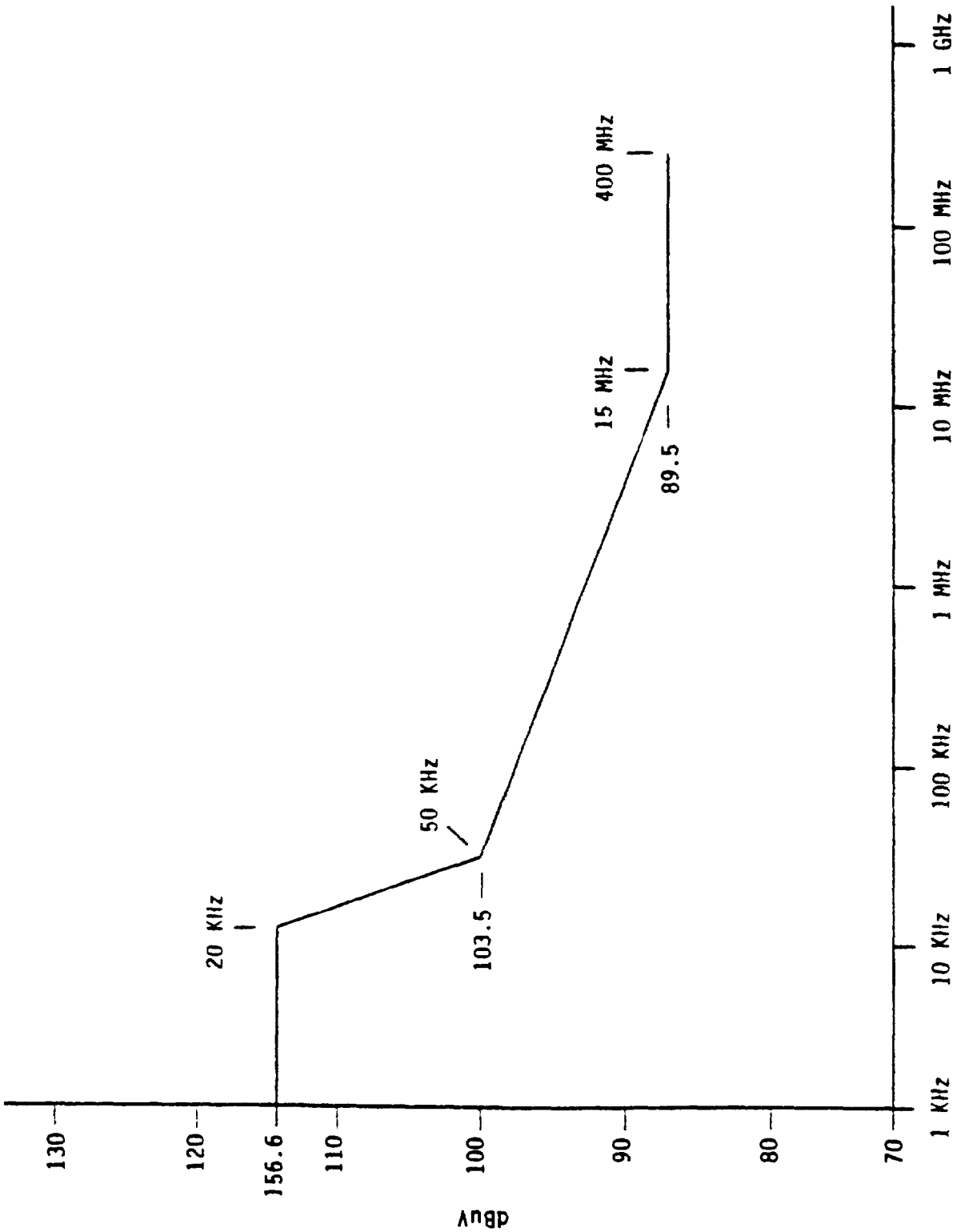
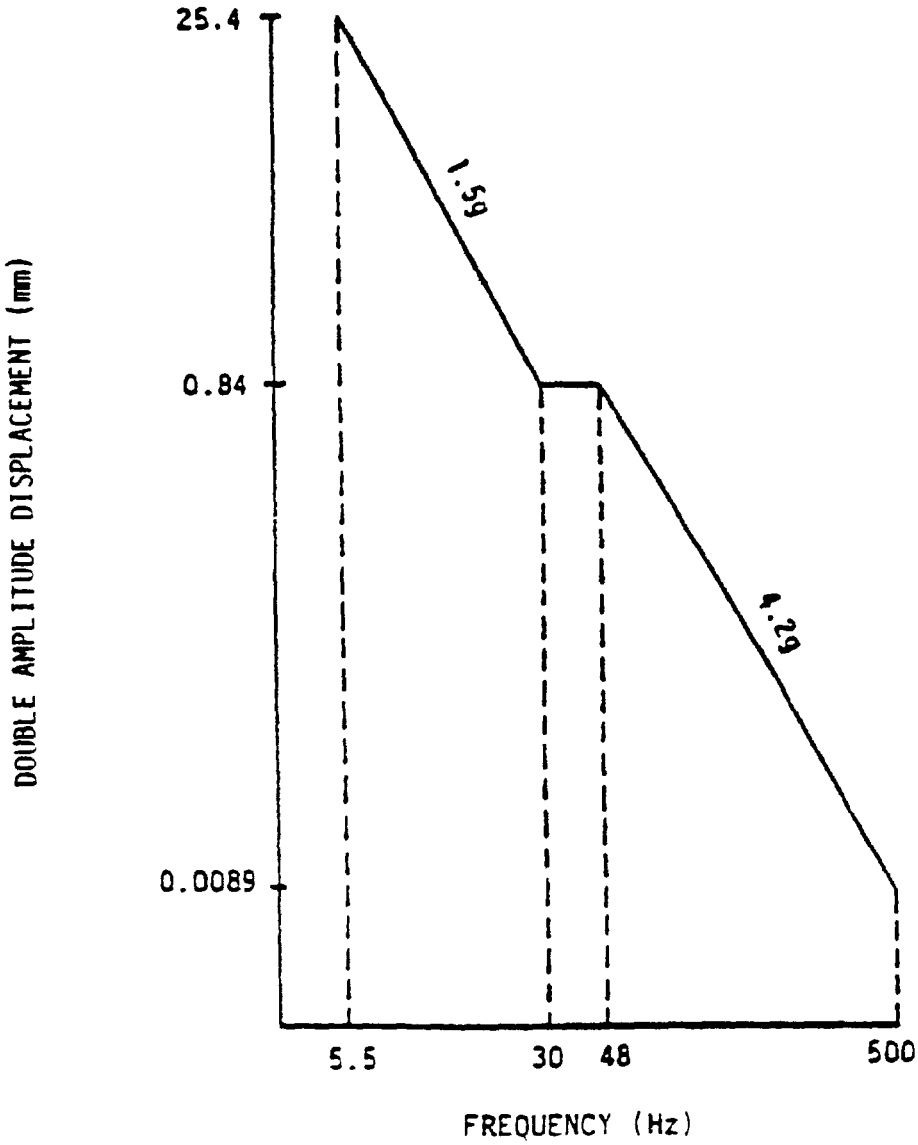


FIGURE 4. Narrowband conducted emissions limit.

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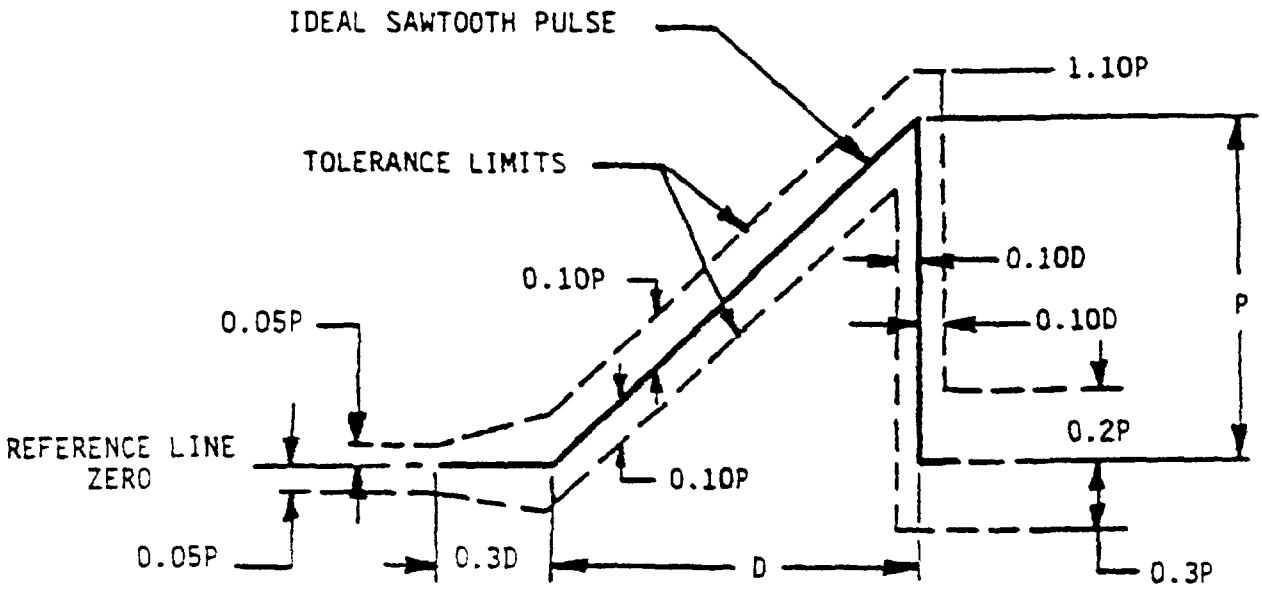


NOTES:

1. Hertz (Hz).
2. Gravity units (g).

FIGURE 5. Vibration test curve.

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NOTES:

1. The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and it's duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of $V_j \pm 0.1\ V_j$, where V_j is the velocity-change associated with the ideal pulse which equals 0.5 PD. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 6. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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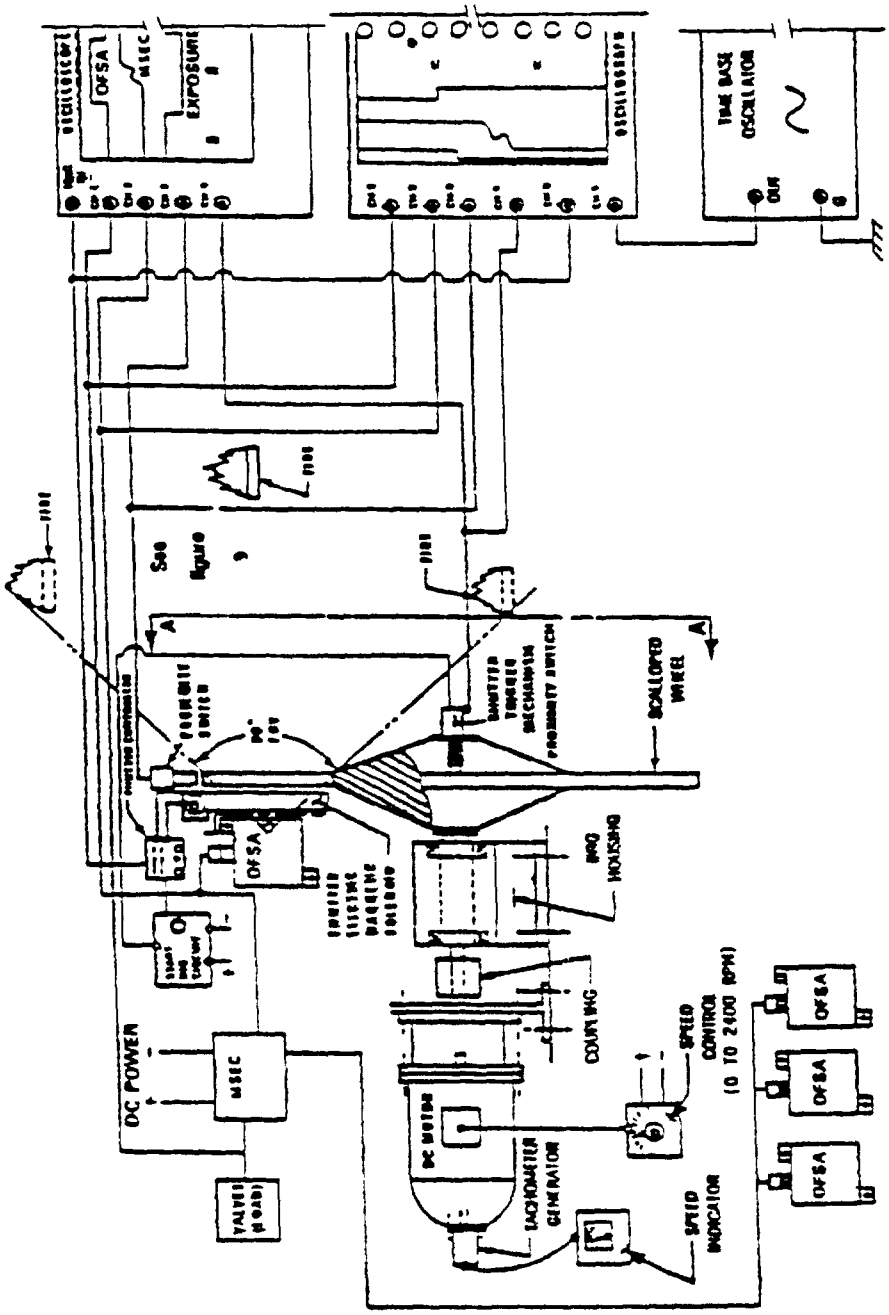
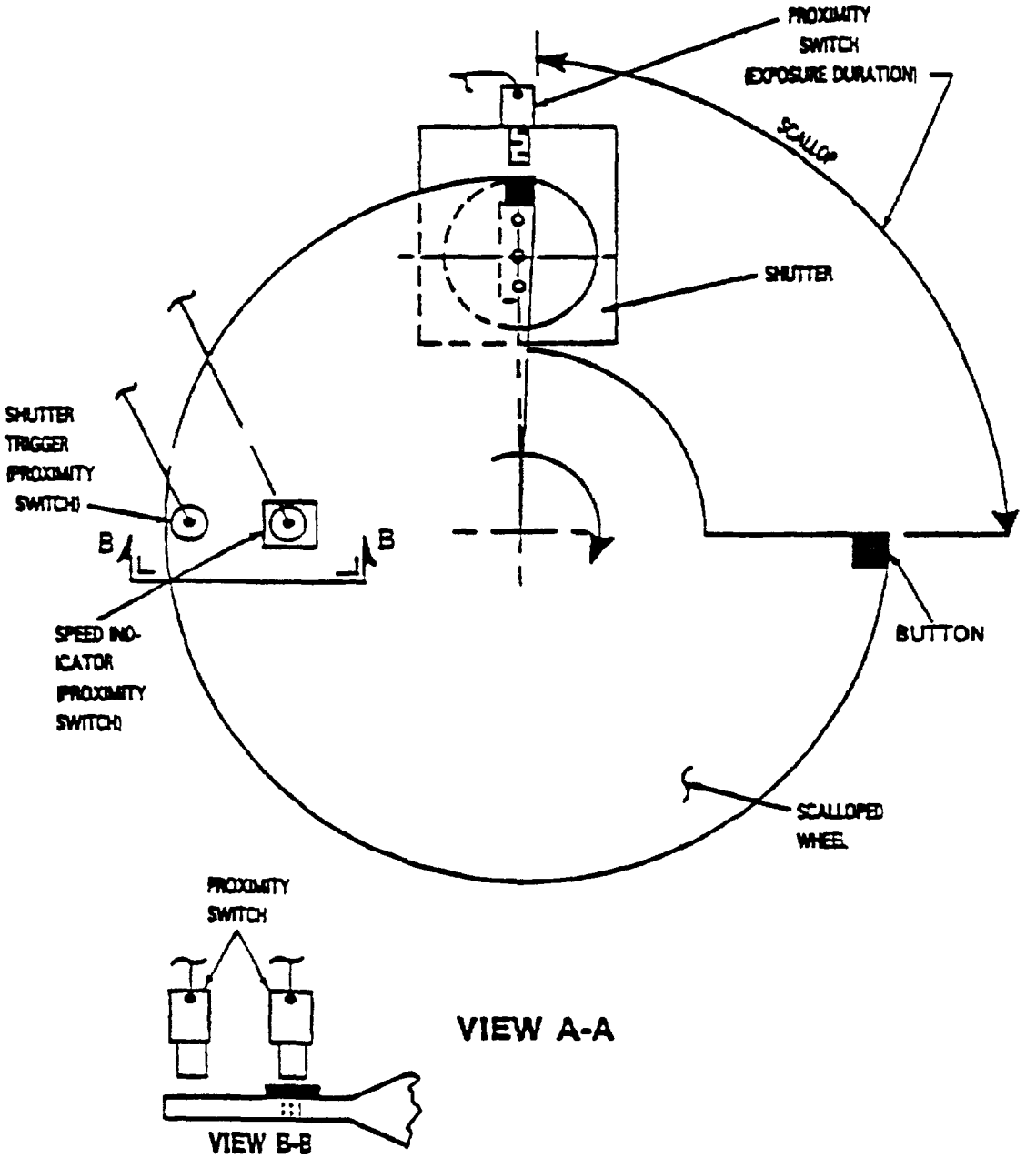


FIGURE 7. Response tests.

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NOTE. All proximity switches shall be made of ferrous material.

FIGURE 8. Response tests.

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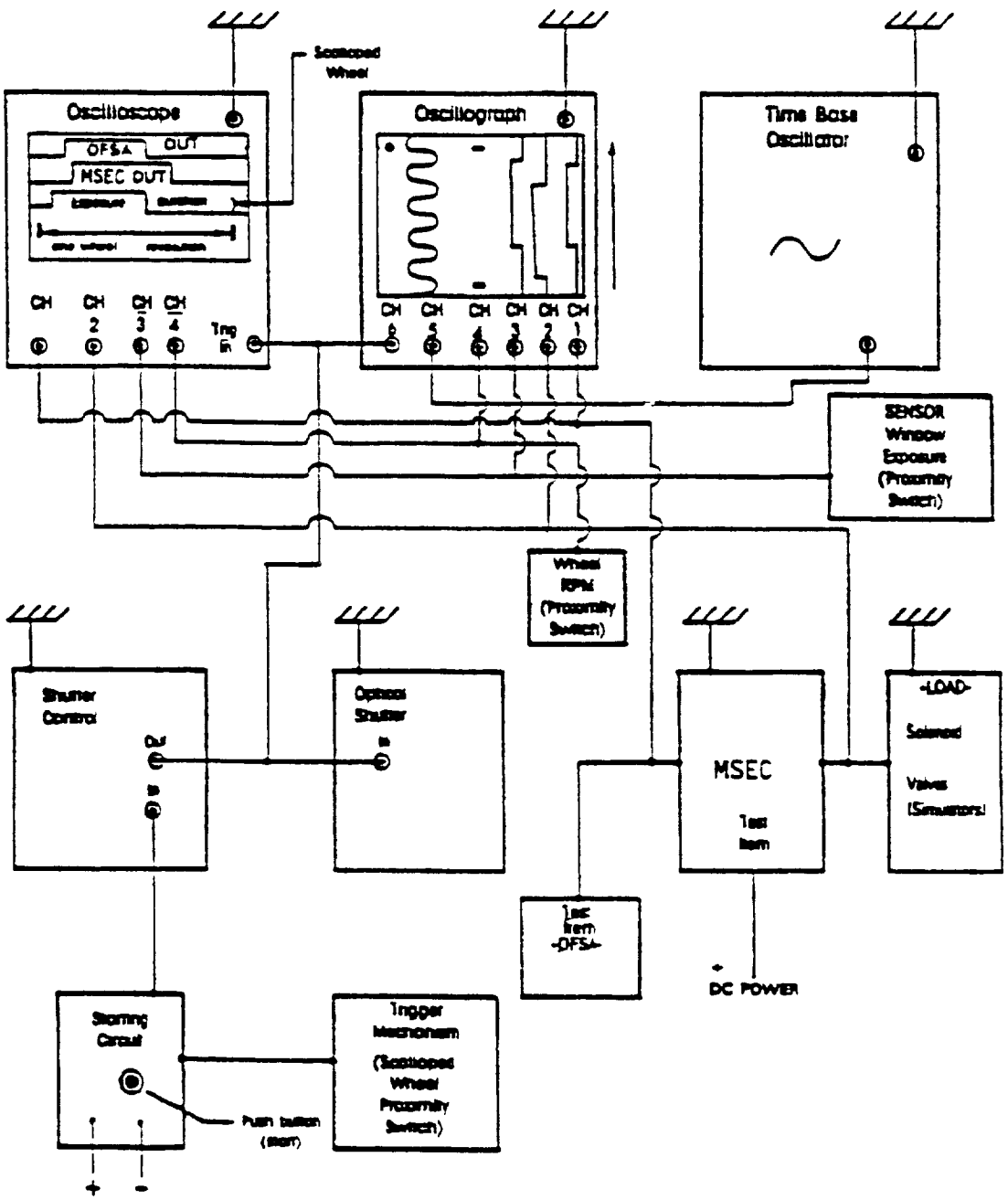


FIGURE 9. Instrumentation and test items schematic for response tests.

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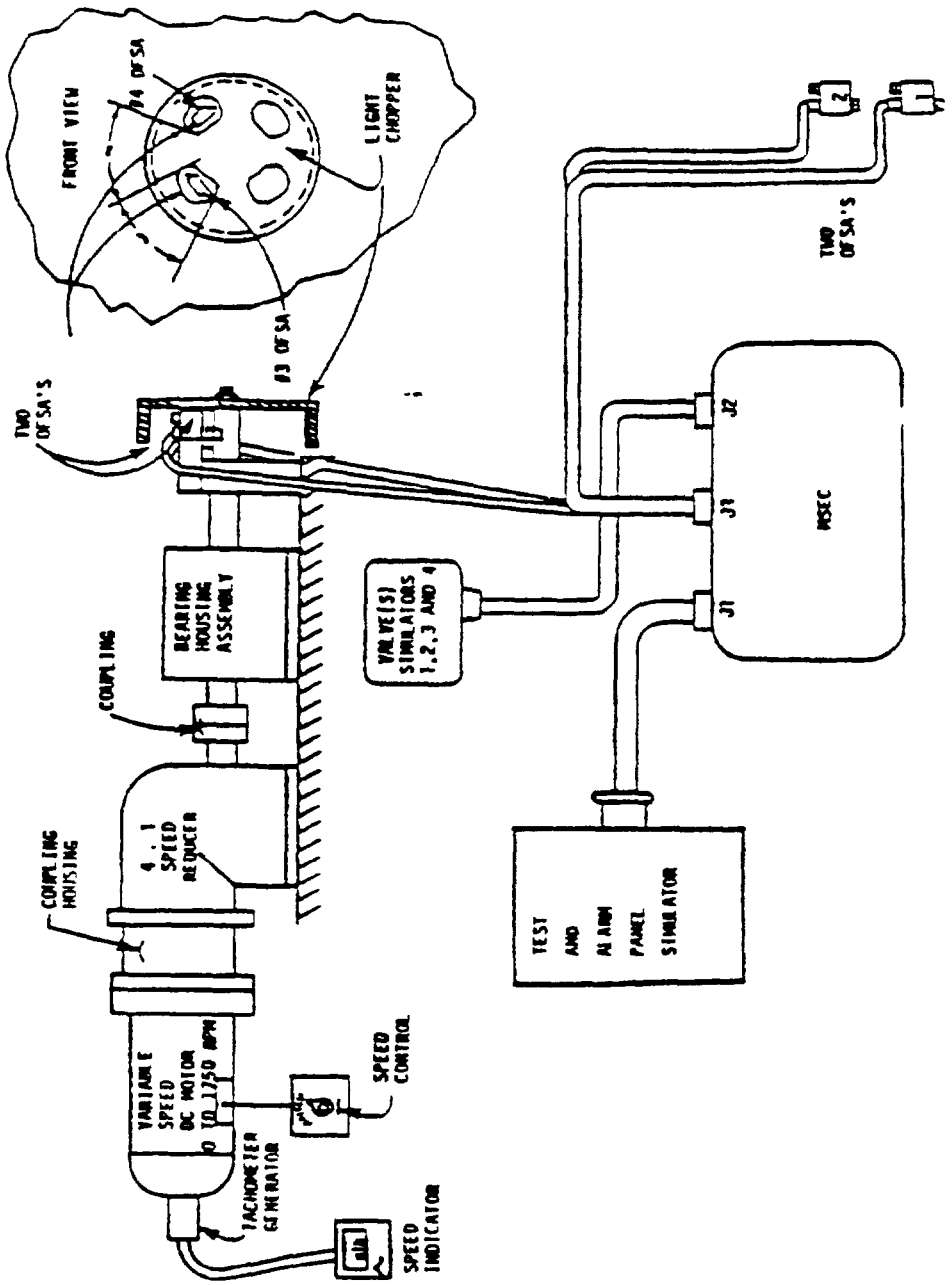


FIGURE 10. False alarm susceptibility tests (chopper light simulator).

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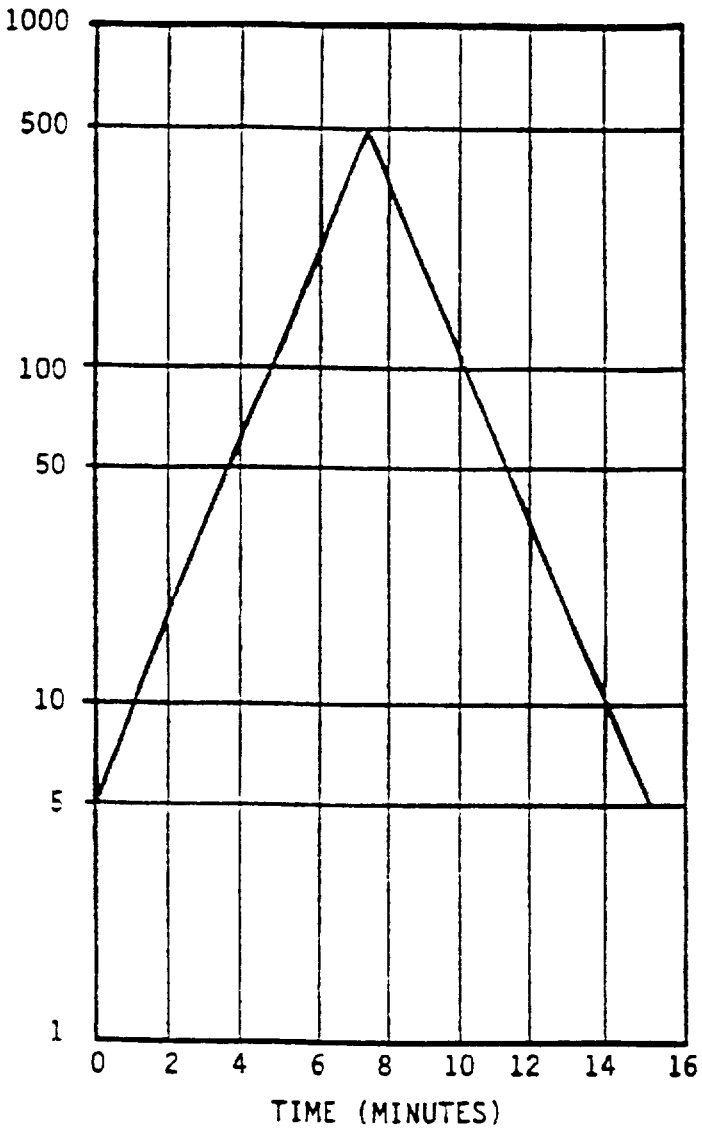
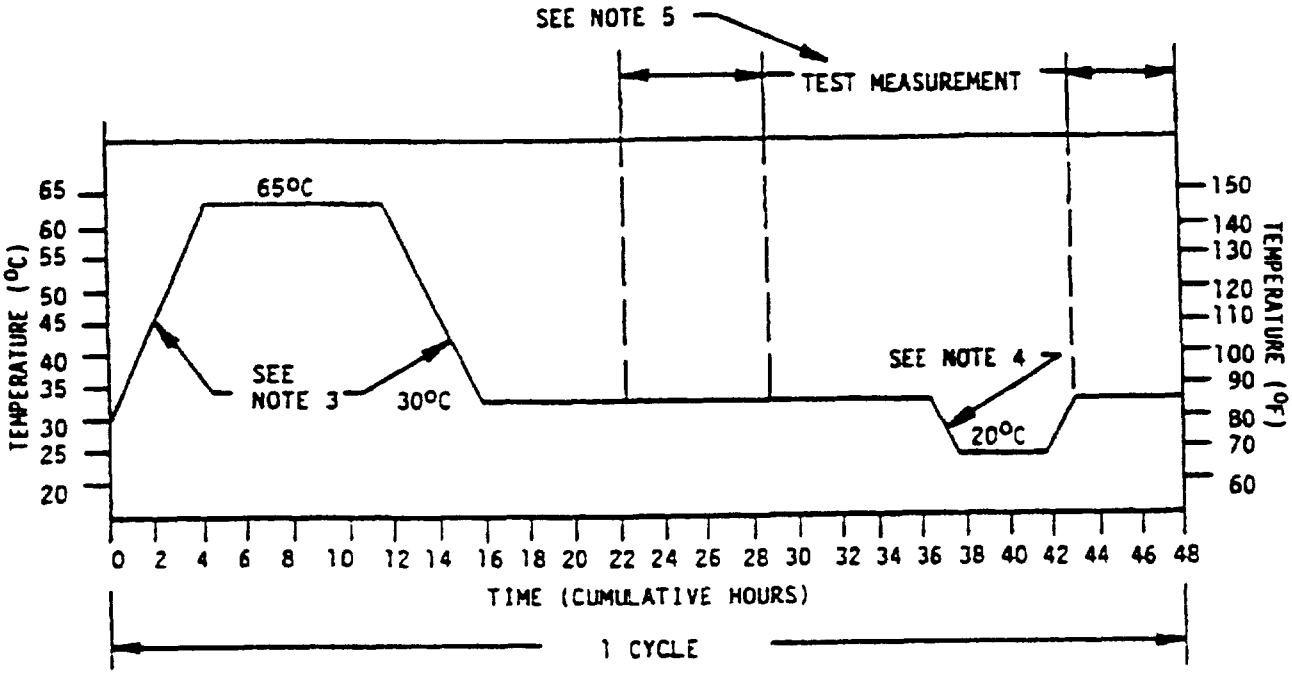


FIGURE 11. Logarithmic sweep.

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NOTES.

1. Tolerance during temperature change shall be not greater than 3°C (5°F).
2. Relative humidity shall be maintained at 94 ± 4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 30 and 65°C (86 and 149°F) shall be not less than 8°C (14.4°F) per hour.
4. The temperature increase in this portion of the curve shall be not less than 10°C (18°F).
5. Test measurements shall be taken only at the period specified in the applicable equipment or system specification.

FIGURE 12. Humidity cycle.

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